IRRATIONALITY MEASURES FOR DIFFERENCES OF TRIGAMMA VALUES

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ABSTRACT. Let $\mu(\xi)$ denote the irrationality exponent of a real number ξ. We prove

$$\mu\Big(G \pm \frac{5\pi^2}{48}\Big) \le 31.2876, \qquad \mu\big(L(2,\chi_{-3})\big) \le 13.7046,$$
$$\mu\Big(L(2,\chi_{-3}) \pm \frac{\pi^2}{9}\Big) \le 9.4998, \qquad \mu(\pi^2) \le 6.5253,$$
$$\mu\Big(\psi'(\tfrac{k}{5}) - \frac{\pi^2}{6}\Big) \le 23.8272 \quad (1 \le k \le 4),$$
$$\mu\Big(\psi'(\tfrac{k}{7}) - \frac{\pi^2}{6}\Big) \le 2207.8697 \quad (1 \le k \le 6).$$

Each of these numbers is, up to a rational affine map, a difference of two values of the trigamma function ψ' at rational points. For $L(2,\chi_{-3})$ and $L(2,\chi_{-3}) \pm \pi^2/9$ the best bound known before was 24781, due to Calegari, Dimitrov and Tang. For π^2 the known bounds are stronger than ours. We know of no earlier irrationality measure for the other numbers. For $\psi'(\frac{k}{5}) - \pi^2/6$ and $\psi'(\frac{k}{7}) - \pi^2/6$ we know of no earlier proof of irrationality.

The approximations are determinants of rectangular Löwner matrices of digamma differences at integer nodes. Their denominators are bounded prime by prime by a finite list of local bounds. A bound at third order in the Smith normal form treats the primes at which the border of the matrix wraps around. Nonvanishing follows from Hermite interpolation in characteristic zero. The size is bounded with Andréief's identity and a lower bound for a discrete logarithmic energy, checked on a finite cover by cells. The proof is formalized in Lean 4.

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Date: October 31, 2026.

2020 *Mathematics Subject Classification.* Primary 11J82; Secondary 11J72, 15A15, 31A15, 68V20.

Key words and phrases. Irrationality exponent, trigamma function, $L(2,\chi_{-3})$, Löwner matrices, Cauchy determinants, Andréief identity, p-adic valuations, Hermite interpolation, formal verification, Lean.

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1. Introduction

1.1. **The result.** The *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the real numbers μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large integers b. It does not change under $\xi \mapsto r\xi + r'$ with $r, r' \in \mathbb{Q}$ and $r \neq 0$. A finite bound for $\mu(\xi)$ is an irrationality measure for ξ, and it shows that ξ is irrational. We prove

$$\begin{aligned} \mu\Big(G \pm \frac{5\pi^2}{48}\Big) &\le 31.2876, \qquad \mu(L(2,\chi_{-3})) \le 13.7046, \\ \mu\Big(L(2,\chi_{-3}) \pm \frac{\pi^2}{9}\Big) &\le 9.4998, \qquad \mu(\pi^2) \le 6.5253, \\ \mu\Big(\psi'(\tfrac{k}{5}) - \frac{\pi^2}{6}\Big) &\le 23.8272 \quad (1 \le k \le 4), \\ \mu\Big(\psi'(\tfrac{k}{7}) - \frac{\pi^2}{6}\Big) &\le 2207.8697 \quad (1 \le k \le 6). \end{aligned} \tag{1.1}$$

Here

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = 0.91596\,55941\ldots,$$
$$L(2,\chi_{-3}) = \sum_{n\ge 0}\Big(\frac{1}{(3n+1)^2} - \frac{1}{(3n+2)^2}\Big) = 0.78130\,24128\ldots$$

are Catalan's constant and the value at 2 of the Dirichlet L-function of the nontrivial character modulo 3.

All these numbers are differences of values of the trigamma function ψ'. For $0 < s < t \le 1$ put

$$\Psi(s,t) = \psi'(s) - \psi'(t) = \sum_{n=0}^{\infty}\Big(\frac{1}{(n+s)^2} - \frac{1}{(n+t)^2}\Big).$$

From $\psi'(\frac14)=\pi^2+8G$, $\psi'(\frac34)=\pi^2-8G$, $\psi'(\frac13)=\frac{2\pi^2}{3}+\frac92 L(2,\chi_{-3})$, $\psi'(\frac23)=\frac{2\pi^2}{3}-\frac92 L(2,\chi_{-3})$, $\psi'(\frac12)=\frac{\pi^2}{2}$ and $\psi'(1)=\frac{\pi^2}{6}$ one gets

$$\Psi(\tfrac14,1)=8\Big(G+\frac{5\pi^2}{48}\Big),\qquad \Psi(\tfrac34,1)=-8\Big(G-\frac{5\pi^2}{48}\Big),$$
$$\Psi(\tfrac13,1)=\frac92\Big(L(2,\chi_{-3})+\frac{\pi^2}{9}\Big),\qquad \Psi(\tfrac23,1)=-\frac92\Big(L(2,\chi_{-3})-\frac{\pi^2}{9}\Big),$$
$$\Psi(\tfrac13,\tfrac23)=9L(2,\chi_{-3}),\qquad \Psi(\tfrac12,1)=\frac{\pi^2}{3}.$$

For $q=5$ and $q=7$ and $\gcd(k,q)=1$ the number $\Psi(\frac kq,1)=\psi'(\frac kq)-\frac{\pi^2}{6}$ is a rational linear combination of π^2 and the values $L(2,\chi)$ for the Dirichlet characters χ modulo q. Indeed

$$\psi'(\tfrac kq)=q^2\sum_{n\equiv k\ (q)}\frac{1}{n^2}=\frac{q^2}{\varphi(q)}\sum_{\chi \bmod q}\bar\chi(k)L(2,\chi).$$

Since μ is invariant under rational affine maps, the bounds (1.1) are the following theorem.

(s,t)	the number, up to a rational affine map	B
$(\frac14,1)$, $(\frac34,1)$	$G+\frac{5\pi^2}{48}$, $G-\frac{5\pi^2}{48}$	31.2876
$(\frac13,\frac23)$	$L(2,\chi_{-3})$	13.7046
$(\frac13,1)$, $(\frac23,1)$	$L(2,\chi_{-3})+\frac{\pi^2}{9}$, $L(2,\chi_{-3})-\frac{\pi^2}{9}$	9.4998
$(\frac k5,1)$, $k=1,2,3,4$	$\psi'(\frac k5)-\frac{\pi^2}{6}$	23.8272
$(\frac k7,1)$, $k=1,\dots,6$	$\psi'(\frac k7)-\frac{\pi^2}{6}$	2207.8697
$(\frac12,1)$	π^2	6.5253

TABLE 1. The pairs of shifts and the exponents of Theorem 1.1.

Theorem 1.1. *Let (s,t) and B be in one row of Table 1. For every $\varepsilon>0$ there is b_0 such that for all coprime integers a, b with $b\ge b_0$,*

$$\Big|\Psi(s,t)-\frac ab\Big|>b^{-B-\varepsilon}.$$

In particular $\Psi(s,t)$ and the number in the same row are irrational, with irrationality exponent at most B.

The rows come from one construction with different shifts. The bound for π^2 is not new. Since $\zeta(2)=\pi^2/6$, irrationality exponents for π^2 and for the value of the Riemann zeta function at 2 coincide. We state the known bounds for π^2. The best known bounds are 5.0193784 of Giraudin [15] and 4.62675 of Leibtag et al. [23]. We include the row because it shows the reach of the construction. One family of determinants, with the same three estimates, covers every row of Table 1.

Catalan's constant itself is a trigamma difference, $G = \frac{1}{16}\Psi(\frac{1}{4}, \frac{3}{4})$. The technique covers this pair of shifts as well. For this pair, however, the content constant given by our local bounds exceeds the decay constant in our computations. So the method gives no irrationality measure for G. The companion paper [24] refines OpenAI's construction for G instead, and proves a finite irrationality exponent for it. The computations for G are not part of the formal proof.

Theorem 1.1 has been verified in the Lean 4 proof assistant, together with every theorem, proposition and lemma used in its proof. The formal proof has no unproved steps and uses only the standard axioms. Section 11 describes it. The remarks and Section 10 discuss the method and are not part of the formal proof. The formalization is available at `https://github.com/slitsey/catalan-irrationality-measure`. The use of AI in this work is disclosed at the end of the paper.

1.2. **Previous results.** *Catalan's constant.* In October 2026 OpenAI [31] proved that G is irrational and formalized the proof in Lean. Sun's preprint [38] had claimed a proof of the irrationality of G. Wachs [41] reports that its main estimate is not justified. Finite irrationality exponents for G were obtained after OpenAI's proof by Jha [17], Carreira [9], Bojnal [4], Xu [42] and the author [24]. Carreira [8] also records attempts to carry the determinant method to $\beta(4)$. For the arithmetic of G and related values before these results see [36, 45, 44, 20, 21] and the survey [48]. Eskandari, Murty and Nemoto [12] construct a 2-dimensional mixed motive over $\mathbb{Q}$ with period G. They use it to obtain linear forms in 1 and G. Eskandari [11] improves the approximation exponent of these forms of Rivoal–Zudilin type from 0.52 to 0.62. These are explicit approximations, not a lower bound for all rationals. Our searches found no irrationality result and no irrationality measure for $G - 5\pi^2/48$ or $G + 5\pi^2/48$.

The value $L(2, \chi_{-3})$. Calegari, Dimitrov and Tang [6] proved that 1, π^2 and $L(2, \chi_{-3})$ are linearly independent over $\mathbb{Q}$. Their proof applies a new arithmetic holonomy bound to a construction of Zagier. In [7] they made holonomy bounds effective. Theorem 5.1 there gives $\mu(L(2, \chi_{-3})) \leq 24781$. The remark that follows that theorem states that the same theorem and proof hold for any $\mathbb{Q}$-linear combination of $L(2, \chi_{-3})$ and π^2. So 24781 is also the known bound for $L(2, \chi_{-3}) \pm \pi^2/9$. Theorem 1.1 improves it to 13.7046 for $L(2, \chi_{-3})$ and to 9.4998 for $L(2, \chi_{-3}) \pm \pi^2/9$. Leibtag et al. [23] find numerically that Apéry-like approximations from conservative matrix fields have positive approximation exponents for $L(2, \chi_{-3}) \pm \pi^2/9$. A supplement to their paper uses one such field to give an Apéry-style proof of the theorem of Calegari, Dimitrov and Tang. The exponent it proves for each of these two numbers is only greater than 10^{-4}.

The numbers $\psi'(\frac{k}{5}) - \pi^2/6$ and $\psi'(\frac{k}{7}) - \pi^2/6$. Our searches found no irrationality result for them. Prellberg [33] proved that $\mathrm{Li}_2(1/5)$ is irrational. That is a different number.

Other constants. Since September 2026 a series of new results has appeared in this area. Several of them were obtained with AI systems. They join classical results such as the bound $\mu(\log 2) < 3.5746$ of Marcovecchio [28], the bounds of Rhin and Viola for π^2 and $\zeta(3)$ [34, 35], which followed Apéry and Beukers [1, 3], and Zudilin's bound for π^2 [46]. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [32]. This improves the bound $\mu(\pi) < 7.1033$ of Zeilberger and Zudilin [43]. Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$ with $r \neq 1$ [25]. Their result also covers nonzero rational combinations of such logarithms. Giraudin lowered the bound for $\mu(\pi^2)$ to 5.0193784 with two hypergeometric constructions of Zudilin [46] at new parameters [15]. Giraudin also lowered the bound for $\mu(\zeta(3))$ to 5.5138800 by applying the Rhin–Viola method at a new integral direction [16]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [23] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields. By searching over other trajectories they obtained $\mu(\pi^2) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [33] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational. These are three cases left open in [6]. The proof uses the integrals of Rhin–Viola and Viola–Zudilin. Fauzan [13] announced a proof that $\zeta(5)$ is irrational, using rationally normalized Hankel determinants. Lai [22] combined that method with simultaneous approximations of Zudilin and of Dauguet and Zudilin. Lai showed that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$.

Determinants. Irrationality proofs built on determinants of large size have a precedent in Zudilin's determinantal criterion with Hankel determinants [47]. The preprint of Liu, Jiang and Zhang on logarithms [25] uses a separated-weight interpolation determinant. The arguments of Fauzan and Lai [13, 22] use Hankel determinants. Brown [5] proposes a multivariate transfinite-diameter test for Mellin-type integrals and applies it to π^2. The framework we use is OpenAI's [31]. In it the decay of the determinants is controlled by Cauchy determinants and Andréief's identity, and the denominators are controlled prime by prime. Our determinants are Löwner matrices, that is, matrices of divided differences, of a digamma difference. They are a matrix-valued analogue of Padé and Hermite–Padé approximation [30]. Formal proofs of irrationality results in Lean also include the irrationality of $\zeta(3)$ by Liu, Zhang and Zhi [26].

1.3. **What is needed.** OpenAI's argument for G and the irrationality measures built on it use one determinant of large size. Its denominators and its decay are balanced very closely. An irrationality measure needs three estimates for a family of determinants $P(x)$ that depend on a variable x. The first bounds the denominators of the coefficients of P. The second bounds $|P(x)|$ at every point x near the constant, not only at the constant. The third shows that some member of the family does not vanish at a given rational point, at a size that can be chosen in terms of the denominator

of that point. In this paper we use a different and much smaller family of determinants. We prove the three estimates once, uniformly in the shift.

1.4. **New ideas.** The proof rests on the following ideas. Section 2 explains each of them in more detail.

(1) *Rectangular frames with an interleaved border.* The matrix has a long common block, which carries the variable, and a border of rows and columns that never meet the diagonal. The degree of the determinant stays that of the common block. The decay is that of a larger configuration of nodes.
(2) *The content from a finite list of local bounds.* Each local bound for the p-adic valuation of the denominator is an explicit integer expression in four integers attached to the prime. A finite table of affine pieces, checked by comparisons of rational numbers, gives the content constant for every shift with the same denominator.
(3) *The content at third order.* Some primes meet a residue class of the border twice. For them Schur complements eliminate the pairs of border nodes in one class. Rational interpolation then separates the remaining border nodes.
(4) *Nonvanishing in characteristic zero.* Hermite interpolation bounds the corank of the matrix at a rational point independently of its size. No prime enters, so the size can be chosen freely.
(5) *Decay through a certificate measure.* A comparison measure, polynomial on finitely many segments, bounds a discrete logarithmic energy from below. The inequalities that remain are checked on a finite cover by cells.

The constants of the proof for each row of Table 1 are listed in Table 2.

1.5. **Organization.** Section 2 explains the construction and the main ideas without technicalities, and states the general deduction. Section 3 gives the construction. Sections 4, 5 and 6 prove the three estimates, content, nonvanishing and decay, uniformly in the shift. Section 7 treats the frame of the pair $(\frac{1}{3}, \frac{2}{3})$. Section 8 deduces Theorem 1.1. Section 9 collects the facts specific to each family, and Section 10 discusses the limits of the method. Section 11 describes the formalization. The disclosure of AI use follows it.

1.6. **Notation and conventions.** We write ψ for the digamma function, the logarithmic derivative of Euler's gamma function, and v_p for the p-adic valuation. For $P \in \mathbb{Q}[x]$, $\operatorname{den} P$ is the least positive integer whose product with P lies in $\mathbb{Z}[x]$. We write $[0, r) = \{0, 1, \dots, r-1\}$. For finite sets X_0, Y_0 of integers and a matrix A indexed by integers, $A[X_0, Y_0]$ is the submatrix with rows X_0 and columns Y_0.

2. Overview of the method

2.1. **Löwner matrices of digamma differences.** Let F be a function analytic near a finite set of distinct points. The *Löwner matrix* [27] of F is

the matrix of divided differences indexed by the points:

$$\mathsf{L}_F[t,t'] = \frac{F(t') - F(t)}{t' - t} \ (t \neq t'), \qquad \mathsf{L}_F[t,t] = F'(t).$$

It is a Cauchy-like structured matrix. Hermite's formula gives $\mathsf{L}_F = \frac{1}{2\pi i} \oint F(z)\, v(z) v(z)^\top dz$ with $v_t(z) = 1/(z-t)$. A representation of F as a Stieltjes or contour integral therefore exhibits L_F as a Gram matrix of Cauchy kernels.

For a rational shift $0 < s < 1$ with denominator q we take the digamma difference F_s. For the pair $(\frac{1}{3}, \frac{2}{3})$ we take a combination F_χ of three digamma functions. The nodes are integers. We put

$$\begin{gathered} F_s(z) = \psi(z+1) - \psi(z+s), \qquad \omega = \tfrac{1}{q^2}, \quad \xi_s = -\tfrac{1}{q^2}\Psi(s,1), \\ F_\chi(z) = -\tfrac{1}{3}\psi(z+\tfrac{1}{3}) + \tfrac{1}{2}\psi(z+\tfrac{1}{2}) - \tfrac{1}{6}\psi(z+1), \\ \omega = -\tfrac{1}{6}, \quad \xi_\chi = \tfrac{1}{36}\Psi(\tfrac{1}{3}, \tfrac{2}{3}). \end{gathered} \tag{2.1}$$

In each case $\xi = \omega\, F'(0)$. The differences $F(t') - F(t)$ and $F'(t) - F'(0)$ are rational for integers $t, t' \geq 0$. Let R and Q be finite sets of nonnegative integer nodes, which index the rows and the columns. Then

$$\omega\, \mathsf{L}_F[R,Q] = M[R,Q] + \xi\, \mathrm{Id}_{R,Q}, \qquad \mathrm{Id}_{R,Q}[t,t'] = \mathbf{1}[t = t'], \tag{2.2}$$

with an explicit rational symmetric kernel M on $\mathbb{Z}^2_{\geq 0}$, given in §3. Replacing ξ by a variable x gives the matrix polynomial $\Phi(x) = M[R,Q] + x\mathrm{Id}_{R,Q}$. Its determinant $P(x)$ has rational coefficients and degree at most $|R \cap Q|$, and $P(\xi) = \det(\omega\, \mathsf{L}_F[R,Q])$.

Suppose that $|\xi - a/b|$ is very small. Then $P(a/b)$ is very close to $P(\xi)$, which is tiny. On the other hand $b^{|R\cap Q|} \operatorname{den}(P)\, P(a/b)$ is an integer. If it is not zero, we get a contradiction as soon as

$$\underbrace{\log \operatorname{den} P}_{\text{content}} + |R \cap Q| \log b \ < \ \underbrace{-\log |P(a/b)|}_{\text{decay}}.$$

Three estimates are therefore needed. The first is an upper bound for the *content* $\log \operatorname{den} P$. The second is an upper bound for $|P(x)|$ near ξ, the *decay*. The third is the *nonvanishing* of $P(a/b)$ for some member of a family of determinants whose size can be chosen in terms of b. Each estimate is proved once, for every shift. The families differ only in a few parameters.

2.2. **Rectangular frames with an interleaved border.** We take a long common block $C = [0, r)$, which carries the variable x. We add a border of $m \approx \alpha r$ row nodes and m column nodes, interleaved:

$$X' = \{r, r+2, \ldots, r+2m-2\}, \qquad Y' = \{r+1, r+3, \ldots, r+2m-1\}.$$

The rows are $R = C \sqcup X'$ and the columns are $Q = C \sqcup Y'$. The determinant $P(x) = \det \Phi(x)$ still has degree at most r. So the integrality loss $|R \cap Q| \log b$ is that of r nodes. The matrix has size $r + m$, and its decay is that of a configuration of $(1+\alpha)r$ nodes. The content grows too, but less, because the border rows and columns never meet the diagonal. The *members* of the

family are the principal restrictions obtained by deleting a bounded number of common nodes. The only parameter of the geometry is $c = 2\alpha$, the length of the border relative to r. We use $c = \frac{3}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{9}, \frac{1}{6}$ for $q = 2, 4, 3, 5, 7$, and $c = \frac{1}{2}$ for F_χ.

The interleaving of X' and Y' is essential. It keeps the two Cauchy determinants in Andréief's formula comparable (§6). It also gives the reduction of the matrix modulo a prime an alternating structure. Several of the arithmetic bounds of §4 rest on this structure.

2.3. **The content from local bounds.** As in the work of Rhin and Viola [34, 35], we bound the p-adic valuation of $\operatorname{den} P$ prime by prime. We then sum the bounds against $\log p$ with the prime number theorem in arithmetic progressions. For a prime $p \asymp r$ the relevant data are the scale r, the border size m, the prime p and the first pole k_p. This is the first index $k_p \geq 1$ at which the second digamma term of F_s has a pole modulo p. We have $k_p = \kappa p + O(1)$ with $\kappa = j/q$, where $j \equiv qs\, p^{-1} \pmod q$ is the *root class* of p. The frame F_χ has two such first poles. On each class of primes modulo 6 both are affine functions of one index of the same kind (Section 7).

We prove a finite list of local bounds, each by its own mechanism. The first is the assignment bound, with explicit potentials. Three bounds concern the rank of the reduction modulo p of a rescaled matrix. Another comes from windows of consecutive border nodes free of poles. In such a window the reduction is translation invariant between residue classes. The window bound also applies to the primes $p \leq m$, which meet the border several times. Modulo p, the border covers an arc of residue classes once more than the others. On this arc the translation argument improves the assignment bound.

Every local bound is an explicit integer expression in (r, m, p, k_p). It is built from affine forms with min, max, truncated subtraction, the distance to $p\mathbb{Z}$ and the multiple $p\lfloor r/p \rfloor$. It holds in a range of p/r described by linear inequalities. Replacing (r, m, p, k_p) by $(1, \alpha, \theta, \kappa\theta)$ turns each bound into a piecewise affine profile in $\theta = p/r$. The error is bounded by an explicit constant. The content constant is then

$$\Lambda = \frac{1}{\varphi(q)} \sum_j \int_0^\infty \mathcal{P}_j(\theta)\, d\theta,$$

where $\mathcal{P}_j$ is the pointwise minimum of the valid profiles for the root class j. A finite table of affine pieces above $\mathcal{P}_j$ is a certificate for Λ. Each piece carries the name of the bound that justifies it. The table is correct if finitely many comparisons of rational numbers hold. One theorem, Theorem 4.14, turns an accepted table into the content estimate. The bounds depend on the shift s only through the root class. So the same table serves every shift with denominator q, and the numbers $\Psi(s, 1)$ for these shifts receive the same exponent.

2.4. **The content at third order.** For the primes $p \le 2m$ the border $[r, r+2m)$ meets some residue classes modulo p more than once. The bounds above work at first or second order in the Smith normal form of the rescaled matrix, and they are not sharp there. We gain at third order. Write $m = np + \bar{m}$ with $0 \le \bar{m} < p$, and $d = \|m\|_p$.

First eliminate the common block. Consider a residue class that carries a pair of border nodes t and $t+p$, one a row and one a column. It contributes a unit pivot. Its leading coefficient is the increment of the frame over $(t, t+p]$. This interval contains exactly one pole of each digamma term, and the two contributions cannot cancel. Eliminating these pivots by Schur complements leaves a matrix whose Smith normal form looks like that of a smaller geometry. This geometry has a shorter common block and a border of d rows and d columns. The separation argument of the bound (S) can be run on it at third order.

Two facts make this work. First, the Schur complement changes the second-order term. The change is exactly compensated by replacing the moments of the common block by the moments of the mixed blocks that the elimination produces. These blocks are not symmetric. The compensation is a pair of identities for bordered inverses. Second, the remaining border nodes must still be separated from each other. For a cut at the second pole we deform the shift, $s \mapsto s + pY$, and evaluate at the pole. There the bordered system becomes a pair of Cauchy blocks. For a cut at a multiple of p we take the leading coefficient in Y, which leaves the harmonic numbers. There the bordered system is a problem of rational interpolation of type (n, n) on consecutive integers. Its value at infinity is $H_{\beta+n} + 2H_n$, where β is the number of common nodes of the class. Two neighbouring classes differ by $1/(\beta+n+1)$, a p-adic unit. The resulting bound (U) is again an explicit integer expression in (r, m, p, k_p). It holds for every member with no loss for the deleted nodes.

2.5. **Nonvanishing.** Let x_0 be a rational point. The kernel of $\Phi(x_0)$ is identified with a space of pairs of polynomials that satisfy Hermite interpolation conditions at the nodes. Two such pairs have vanishing cross product, by a count of degrees and zeros. A further pair in the kernel would give a rational antidifference of the increment of the frame. No such antidifference exists, because the pole at -1 is isolated in its class modulo $\mathbb{Z}$. This bounds the corank of $\Phi(x_0)$ by a number that depends only on the frame and the period of the geometry. A principal restriction that deletes at most that many common nodes is then nonsingular. This is Theorem 5.1. No prime enters, so the size r can be chosen freely in terms of b.

2.6. **Decay.** Moving Hermite's contour to a polygonal path Γ in the left half-plane writes $\omega \llcorner_F [R, Q]$ as a Gram matrix $\int v_t v_{t'} \, d\Omega$. Here Ω is a complex measure. It has signed atoms at poles of F and an exponentially small density on Γ. Andréief's identity with two families of functions, one for the

rows and one for the columns, gives

$$|\det \mathsf{L}_F[R,Q]| \le \frac{1}{n_*!}\int |\mathsf{C}_R(z)|\,|\mathsf{C}_Q(z)| \prod_{i=1}^{n_*} d|\Omega|(z_i),$$

where n_* is the size of the matrix and C_R, C_Q are Cauchy determinants. Taking logarithms turns the integrand into a discrete logarithmic energy. The free points z_i interact with the nodes in an external field.

We bound this energy from below by comparing it with a *certificate measure* σ. This is a density that is polynomial on each of finitely many segments. The comparison uses the conditional positive definiteness of the logarithmic kernel. What is left is a lower bound for one explicit function Υ. It is needed on the contour, on the segment carrying the atoms and on the interval of the nodes. We prove it on a finite cover of these sets by cells, each checked in exact rational arithmetic. One theorem, Theorem 6.2, gives the decay for every geometry parameter $c \le 1$ and every frame whose poles are separated from the contour. A family supplies only σ, the polygon and the cover.

2.7. **The deduction.** Write $\eta = D - \Lambda$, where D is the decay constant. Let γ be the window constant of the decay estimate. This means that the decay holds at every point within $e^{-\gamma r}$ of ξ. Section 8 combines the three estimates. The result is

$$\mu(\xi) \le \frac{\gamma}{\eta} = \frac{\gamma}{D-\Lambda}.$$

2.8. **Formal verification.** Every step is formalized in Lean 4 with Mathlib [10, 39]. This includes the local arithmetic bounds and the checker of the content tables. It includes the interpolation argument for nonvanishing, the contour representation and the cell checks of the decay certificates. It also includes the prime number theorem in arithmetic progressions. The formal proofs are built on a shared core. The core contains rectangular determinants, Gauss valuations, the local bounds, the content checker, the selection in characteristic zero, the decay theorem and the measure deduction. A family supplies its frame, its geometry parameter and its data tables. The written proofs below are at the level of a research article. For long computations we point to the formal proof, which is the basis for the correctness of Theorem 1.1.

3. The construction

3.1. **The kernels.** Let $0 < s < 1$ be rational with denominator q. Put $o = q(1-s)$, so that $k - 1 + s = (qk - o)/q$. For the function F_s of (2.1), the kernel M of (2.2) is as follows. Here $0 \le t < t'$ are integers, and

$M[t',t] = M[t,t']$.

$$M[t,t'] = \frac{1}{q^2(t'-t)} \sum_{k=t+1}^{t'} \Big(\frac{1}{k} - \frac{q}{qk-o}\Big),$$

$$M[t,t] = -\frac{1}{q^2} \sum_{k=1}^{t} \Big(\frac{1}{k^2} - \frac{q^2}{(qk-o)^2}\Big).$$

For $s = \frac{3}{4}$ this is $M[t,t'] = -\frac{1}{t'-t} \sum_{k=t+1}^{t'} \frac{1}{16k(4k-1)}$. For the function F_χ,

$$M[t,t'] = \frac{1}{6(t'-t)} \sum_{k=t+1}^{t'} g(k), \qquad M[t,t] = \frac{1}{6} \sum_{k=1}^{t} g'(k),$$

with $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$. Then $-\frac{1}{6}g'(k) = \frac{1}{2(3k-2)^2} - \frac{1}{3(2k-1)^2} + \frac{1}{36k^2}$.

We write each frame as $F = \sum_{\varsigma\in\mathcal{T}} w_\varsigma \psi(z+\varsigma)$, where $\mathcal{T}$ is the set of shifts of F and $\sum_\varsigma w_\varsigma = 0$. For F_s, $\mathcal{T} = \{1, s\}$ with $w_1 = 1$ and $w_s = -1$. For F_χ, $\mathcal{T} = \{\frac{1}{3}, \frac{1}{2}, 1\}$. The number $|\mathcal{T}|$ of digamma terms is 2 for F_s and 3 for F_χ.

Lemma 3.1. *For each function of (2.1), $\omega \mathsf{L}_F[R,Q] = M[R,Q] + \xi \mathrm{Id}_{R,Q}$ for all finite sets $R, Q \subset \mathbb{Z}_{\ge 0}$, with ω and ξ as in (2.1).*

Proof. From $\psi(w+1) - \psi(w) = 1/w$ one gets $F_s(k) - F_s(k-1) = \frac{1}{k} - \frac{1}{k-1+s} = \frac{1}{k} - \frac{q}{qk-o}$. Telescoping gives the off-diagonal entries. In the same way $\psi'(w+1) - \psi'(w) = -1/w^2$ gives $F_s'(t) - F_s'(0)$. Finally $\omega F_s'(0) = q^{-2}(\psi'(1) - \psi'(s)) = -q^{-2}\Psi(s,1) = \xi_s$. For F_χ the entries are computed in the same way. For the value at 0, the multiplication formula $\sum_{i<n} \psi'(z + \frac{i}{n}) = n^2 \psi'(nz)$ gives $\psi'(\frac{1}{2}) = 3\psi'(1)$ and $\psi'(\frac{1}{3}) + \psi'(\frac{2}{3}) = 8\psi'(1)$. Hence $F_\chi'(0) = -\frac{1}{3}\psi'(\frac{1}{3}) + \frac{1}{2}\psi'(\frac{1}{2}) - \frac{1}{6}\psi'(1) = -\frac{1}{6}\Psi(\frac{1}{3}, \frac{2}{3})$, and $-\frac{1}{6}F_\chi'(0) = \frac{1}{36}\Psi(\frac{1}{3}, \frac{2}{3}) = \xi_\chi$. □

Since $\mu(u\xi + u') = \mu(\xi)$ for rational $u \neq 0$ and u', Theorem 1.1 is a statement about the numbers ξ of (2.1).

3.2. **Geometries.**

Definition 3.2. Let $r, m \ge 0$. The *geometry* $\mathcal{G}(r,m)$ consists of the common nodes $C = [0,r)$, the row nodes $X' = \{r + 2i : 0 \le i < m\}$ and the column nodes $Y' = \{r + 2i + 1 : 0 \le i < m\}$. Rows are indexed by $R = C \sqcup X'$ and columns by $Q = C \sqcup Y'$. Let $I \subseteq C$ be a set of deleted common nodes. The matrix

$$\Phi_{\mathcal{G},I}(x) = \big(M[t,t'] + x\,\mathbf{1}[t = t' \in C \setminus I]\big)_{t\in R\setminus I,\ t'\in Q\setminus I}$$

is square of size $|C \setminus I| + m$. Put $P_{\mathcal{G},I}(x) = \det \Phi_{\mathcal{G},I}(x) \in \mathbb{Q}[x]$. We write $\Phi_\mathcal{G}$ and $P_\mathcal{G}$ when $I = \emptyset$.

By Lemma 3.1, $\Phi_\mathcal{G}(\xi + h) = \omega \mathsf{L}_F[R,Q] + h\mathrm{Id}_{R,Q}$ for real h. The geometry with $m = 0$ is the square Löwner matrix of size r.

Definition 3.3. Let $0 < c \le 1$ and $\alpha = c/2$. The *alternating geometry* of parameter c at an even scale r is $\mathcal{G}(r, m(r))$ with $m(r) = \lceil \alpha r - \frac{1}{2} \rceil$. Then $|m(r) - \alpha r| \le \frac{1}{2}$, and the border $[r, r + 2m)$ has length $cr + O(1)$. For a depth $K \ge 0$, its *members* are the matrices $\Phi_{r,I} = \Phi_{\mathcal{G},I}$ with $|I| \le K$. We put $P_{r,I} = \det \Phi_{r,I}$.

For the frames F_s we use $c = \frac{3}{10}, \frac{1}{5}, \frac{3}{10}, \frac{2}{9}, \frac{1}{6}$ for $q = 2, 4, 3, 5, 7$. For F_χ we use $c = \frac{1}{2}$, as explained in Section 7. All $o(1)$ terms below are uniform over the members as $r \to \infty$.

Lemma 3.4 (integrality)**.** *$P_{\mathcal{G},I}$ has degree at most $|C \setminus I|$. If a/b is a fraction in lowest terms with $P_{\mathcal{G},I}(a/b) \ne 0$, then $|\operatorname{den}(P_{\mathcal{G},I})\, P_{\mathcal{G},I}(a/b)| \ge b^{-|C\setminus I|}$.*

Proof. The variable occurs only in the $|C \setminus I|$ diagonal entries indexed by $C \setminus I$. So each Leibniz term has degree at most $|C \setminus I|$. Then $b^{|C\setminus I|} \operatorname{den}(P_{\mathcal{G},I})\, P_{\mathcal{G},I}(a/b)$ is a nonzero integer. □

Remark 3.5. Suppose that the $m \times m$ block $M[X', Y']$ is invertible. The Schur complement then gives $P_{\mathcal{G}}(x) = \det M[X', Y'] \cdot \det(x\mathrm{Id}_C + \widetilde{M})$ with $\widetilde{M} = M[C, C] - M[C, Y']M[X', Y']^{-1}M[X', C]$. So the rectangular determinant is, up to a constant, the characteristic polynomial of a rational matrix of size $|C|$, as in the square case. We do not need this factorisation. Lemma 3.4 only uses the degree bound.

4. Arithmetic: the content

In this section we bound the common denominator of the coefficients of the members $P_{r,I}$ for the frames F_s. The method is that of Rhin and Viola [34, 35]. The p-adic valuation of the denominator is bounded prime by prime. The bounds are summed against $\log p$ with the prime number theorem in arithmetic progressions. What is new is the form of the local bounds. Each is proved by its own argument. All of them are explicit integer expressions in four integers attached to the prime. Their minimum, integrated, gives the content constant. The comparison between the local bounds and a table of affine pieces is a finite computation with rational numbers. Theorem 4.14 turns a table that passes it into the content estimate. Section 7 treats the frame F_χ.

4.1. The root class of a prime.

Fix the frame F_s, with s of denominator q, and the alternating geometry of parameter $c = 2\alpha$. Put $N = r + 2m$. Let $p > q$ be a prime. The indices $k \ge 1$ at which the entries of M acquire a denominator divisible by p are the multiples of p and the solutions of $qk \equiv o \pmod p$. Let k_p be the least of the latter. Then

$$qk_p = jp + o, \qquad 0 < j < q, \qquad j \equiv qs\, p^{-1} \pmod q.$$

We call j the *root class* of p and put $\kappa = \kappa_j = j/q$. Then $k_p - \kappa p = o/q \in [\frac{1}{q}, 1 - \frac{1}{q}]$. So the first pole is at distance less than one from κp, uniformly in s. Throughout we assume $q(r + 2m) < p^2$. Then every pole below N is simple,

and only the first power of p matters. The primes with $p^2 \le q(r+2m)$ and the prime powers contribute $O(r^{3/2}\log r)$ in total.

4.2. **Gauss valuations, the assignment bound and the rank of the reduction.** Fix a prime p. Let $\mathcal{O} \subset \mathbb{Q}(x)$ be the localisation of $\mathbb{Z}_{(p)}[x]$ at the prime ideal (p). It is a discrete valuation ring with uniformiser p and residue field $\mathbb{F}_p(x)$. Its valuation v, restricted to $\mathbb{Q}[x]$, is the *Gauss valuation* $v(f) = \min_k v_p(f_k)$ for $f = \sum f_k x^k$. By Gauss's lemma $v(fg) = v(f) + v(g)$, and $v_p(\operatorname{den} f) = \max(0, -v(f))$. For a member we write $e_p(r, I) = v_p(\operatorname{den} P_{r,I})$.

Lemma 4.1 (assignment bound and LP duality)**.** *Let A be a square matrix over $\mathbb{Q}[x]$ with rows X_0 and columns Y_0. Let $W_{tt'} = \max(0, -v(A_{tt'}))$, and let $\tau(W)$ be the largest weight $\sum W_{tt'}$ of a perfect matching between X_0 and Y_0. Then $v(\det A) \ge -\tau(W)$. If real numbers u_t, $u'_{t'}$ satisfy $W_{tt'} \le u_t + u'_{t'}$ for all $t \in X_0$ and $t' \in Y_0$, then $\tau(W) \le \sum_t u_t + \sum_{t'} u'_{t'}$.*

Proof. Each Leibniz term is a product of entries along a perfect matching. So its valuation is at least minus the weight of the matching. The valuation of a sum is at least the minimum of the valuations of its terms. The second statement is weak LP duality for the assignment problem. The weight of a perfect matching is at most $\sum_t u_t + \sum_{t'} u'_{t'}$. □

Lemma 4.2 (rank of the reduction)**.** *Let (u, u') be integer potentials with $W_{tt'} \le u_t + u'_{t'}$ for the matrix $\Phi_{r,I}$ at p, of value $\tau' = \sum_t u_t + \sum_{t'} u'_{t'}$. Then $\widetilde{\Phi} = \operatorname{diag}(p^u)\,\Phi_{r,I}\,\operatorname{diag}(p^{u'})$ has entries in $\mathcal{O}$, and*

$$e_p(r, I) \le \tau' - \operatorname{corank}_{\mathbb{F}_p(x)}(\widetilde{\Phi} \bmod p).$$

Proof. We have $v(\det \Phi_{r,I}) = -\tau' + v(\det \widetilde{\Phi})$. Over the discrete valuation ring $\mathcal{O}$ the Smith normal form shows $v(\det \widetilde{\Phi}) \ge \operatorname{corank}(\widetilde{\Phi} \bmod p)$. □

The rank bounds below use a refinement of Lemma 4.2 at second order. Suppose that, after a unimodular change of rows and columns over $\mathcal{O}$, the first n_1 rows and the first n_2 columns of $\widetilde{\Phi}$ are divisible by p. Suppose also that the $n_1 \times n_2$ block where they meet is divisible by p^2. Dividing the first n_1 rows by p leaves the first n_2 columns divisible by p. So

$$v(\det \widetilde{\Phi}) \ge n_1 + n_2. \tag{4.1}$$

The vectors that produce such rows and columns are kernel vectors of the reduction of $\widetilde{\Phi}$ on single residue classes. The condition on the corner is a statement about their pairing at first order, which we now describe. Let $\widetilde{\Phi} = p^2\Phi_{r,I}$, so that all potentials equal 1. Since $q(r+2m) < p^2$, two distinct nodes $t \ne t'$ have $v_p(t'-t) \ge 1$ only if $t \equiv t' \pmod p$, and then $v_p(t'-t) = 1$. Hence $\widetilde{\Phi}$ has entries in $\mathcal{O}$. Its reduction is block diagonal over the residue classes modulo p, which we call the *fibres*. For nodes t, t' in distinct fibres,

$$\widetilde{\Phi}_{tt'} \equiv \frac{p\,(V(t') - V(t))}{t' - t} \pmod{p^2}. \tag{4.2}$$

Here $V(t) \in \mathbb{F}_p$ is the residue of p times the partial sum $\sum_{k \le t}(\frac{1}{k} - \frac{q}{qk-o})$, up to the normalising factor q^{-2}. The function V is a step function of t which jumps only at the poles $t \equiv 0$ and $t \equiv k_p$. Since $t' - t$ is prime to p, only its residue matters in (4.2).

For a vector $\mathbf{y}$ indexed by nodes, its *moment pair* is

$$\hat{\mathbf{y}} = \Big(\sum_t y_t,\ \sum_t y_t V(t)\Big).$$

Lemma 4.3 (pairing)**.** *Let $\mathbf{y}$ be supported on the rows of one fibre and $\mathbf{y}'$ on the columns of a different fibre. Let t and t' be nodes of the first and of the second fibre. Then*

$$\mathbf{y}^\top \widetilde{\Phi}\, \mathbf{y}' \equiv \frac{p}{t' - t} \det \begin{pmatrix} \hat{\mathbf{y}} \\ \hat{\mathbf{y}}' \end{pmatrix} \pmod{p^2}.$$

In particular the pairing vanishes modulo p^2 whenever the moment pairs of $\mathbf{y}$ and $\mathbf{y}'$ are proportional.

Proof. Expand $\mathbf{y}^\top \widetilde{\Phi}\, \mathbf{y}'$ with (4.2). □

The moment pairs are never both zero. This follows from the displacement structure of Löwner matrices.

Lemma 4.4 (moment lemma)**.** *Let A be a matrix with rows indexed by a set X_0 of distinct integers and columns by a set Y_0, with $|X_0| = |Y_0| + 1$. Suppose that its off-diagonal entries are $A(t, t') = (\mathcal{L}(t') - \mathcal{L}(t))/(t' - t)$ for some function $\mathcal{L}$, and that its diagonal entries, on $X_0 \cap Y_0$, are arbitrary. Then A has a nonzero left kernel vector $\mathbf{y}$ with $(\sum_t y_t, \sum_t y_t \mathcal{L}(t)) \neq (0, 0)$.*

Proof. Write $\operatorname{diag}(X_0)$ and $\operatorname{diag}(Y_0)$ for the diagonal matrices of the indices. Then $\operatorname{diag}(X_0)A - A\operatorname{diag}(Y_0) = \mathcal{L}_{X_0}\mathbf{1}^\top - \mathbf{1}\mathcal{L}_{Y_0}^\top$, because the diagonal entries cancel on common indices. The left kernel $\mathcal{V}$ of A is nonzero. Suppose that every $\mathbf{y} \in \mathcal{V}$ had zero moment pair. Multiplying on the left by $\mathbf{y}^\top$ would give $\mathbf{y}^\top \operatorname{diag}(X_0)A = 0$. So $\operatorname{diag}(X_0)$ would preserve $\mathcal{V}$. As this matrix is diagonal with distinct entries, $\mathcal{V}$ would contain a standard basis vector. That vector is supported on one index t, and its moment pair is $(1, \mathcal{L}(t)) \neq (0, 0)$. This is a contradiction. □

4.3. **The local bounds.** We now list the local bounds. Each is a statement of the following form. Let $P_{r,I}$ be a member with $|I| \le K$, and let $p > p_0$ be a prime with $q(r + 2m) < p^2$ that satisfies the stated conditions. Then

$$e_p(r, I) \le \tau(r, m, p, k_p) + \text{debit} \cdot |I|. \tag{4.3}$$

Here τ is the stated expression, and the *debit*, 0 or 1, is the stated cost of each deleted node. The constant p_0 depends only on the frame. We write $(\cdot)_+ = \max(\cdot, 0)$, $\|m\|_p$ for the distance of m to $p\mathbb{Z}$, and $\bar{u} = (k_p + p - r)_+$.

Theorem 4.5 (local bounds)**.** *The following bounds of the form (4.3) hold.*

	bound τ	*conditions*	*debit*
(T1)	$2(r+m)$	*none*	0
(T1′)	$2r+2m-\|m\|_p$	*none*	0
(T2)	$2(r-\bar{u})_+ + 2m + \min(\bar{u}, 2m)$	$2m < p \le r$	0
(T3)	$2r+m$	$2m < p \le r$	0
(T4)	$2(r-k_p)_+ + 2m + (2m-(p-r))_+$	$r < p$	0
(T5)	$(2m-(k_p-r)_+)_+$	$N \le p$, $r \le k_p$	0
(LP1)	$\tau_{\mathrm{T2}} - (\min(\bar{u}, 2m) - m)_+$	$p \le r < p+k_p$, $N \le 2p$	1
(LP2)	$2(r-k_p)+3m$	$r < p < r+m$, $k_p \le r$, $N \le 2p$	1
(W)	$2r+2m-(t_2-t_1)$ *for each window* $[t_1,t_2)$ *of Theorem 4.7*	*as there*	0
(R)	Π_R *of* (4.4)	$2m < p$	0
(S)	Π_S *of* (4.5)	$2m \le p$	1
(U_n)	$\Pi_{U,n}$ *of* (4.6), $n \ge 0$	$np \le m < (n+1)p$, *and* $p \le 2m$ *if* $n = 0$	0

The rest of this subsection proves Theorem 4.5, family by family.

4.3.1. *The assignment bounds (T1)–(T5).* These follow from Lemma 4.1 with explicit potentials. Consider an off-diagonal entry $M[t,t']$. By the ultrametric inequality, $-v_p(M[t,t'])$ is at most the number of the following two events that occur: the interval $(t,t']$ contains a pole, and p divides $t'-t$. A diagonal entry $M[t,t]$ has $-v_p \le 2$ if $t \ge k_p$ and $-v_p \le 0$ if $t < k_p$. The bound (T1) is the trivial bound, with potentials 1 on every row and column.

For (T1′) the potentials depend on the residue class modulo p. On the rows of a class they are $\frac{1}{2}$, $\frac{3}{2}$ or 1, according as X' has more, fewer or as many nodes in the class as Y'. On the columns they are $\frac{3}{2}$, $\frac{1}{2}$ or 1 in the same three cases. The border alternates between X' and Y'. So the imbalance is at most one per class, and the number of unbalanced classes is $2\|m\|_p$.

For (T2)–(T5) one starts from the potential $\mathbf{1}[t \ge k_p]$ on the common nodes t, which pays for the diagonal. One adds 1, or $1+\mathbf{1}[t \ge p]$, on the border nodes t. Then one raises the potentials of the common nodes that have a congruent partner in the border. In each range one checks $W \le u+u'$ entrywise. Two facts are used. Two distinct nodes not congruent modulo p have weight at most 1. Two border nodes are never congruent once $2m < p$. For (T5) we have $N \le p$ and $r \le k_p$. Then no common node pays for the diagonal, and only the border nodes beyond k_p carry potential 1. So (T5) vanishes once $k_p \ge N$, which happens for $p/r > \max(1+2\alpha, 1/\kappa) + o(1)$. All cases are checked in the formal proof.

4.3.2. *Primes near r: the bounds (LP1) and (LP2).* For p close to r the reduction of the rescaled matrix has a large block of rank one per residue class. Its corank can be computed. We give the proof of (LP2). The bound (LP1) applies to the primes with $p \le r < p + k_p$ and $N \le 2p$. It is proved by the same method. With the potentials of (T2), eliminating an invertible block of common nodes brings the reduction to the form $\left(\begin{smallmatrix} \tilde{x}\mathrm{Id} & \mathcal{J}_Y \\ \mathcal{J}_X & 0 \end{smallmatrix}\right)$, where $\tilde{x}$ is the variable x plus a constant. In this form the border nodes below $p+k_p$ contribute rows and columns proportional to unit vectors. The Frobenius–König theorem bounds its rank. The saving over (T2) is $(\min(2m, p + k_p - r) - m)_+$, where $\min(2m, p + k_p - r)$ is the number of border nodes below $p + k_p$. We refer to the formal proof for the details.

Theorem 4.6. *Assume $r < p < r + m$, $k = k_p \le r$ and $N = r + 2m \le 2p$, and that every pole below N is simple. Then $v(\det \Phi_{r,\emptyset}) \ge -(2(r-k)+3m)$. The same holds for every member, with debit 1 per deleted node.*

Proof. Use the potentials $u(t) = \mathbf{1}[k \le t]$ for common nodes $t < r$ and $u(t) = 1 + \mathbf{1}[p \le t]$ for border nodes t. Their value is $\tau' = 2(r-k) + 2m + (N-p)$. Since $N \le 2p$, the poles below N are k, p, and $p + k$ if $p + k < N$. Each is simple. The residues of p times the increment $\frac{1}{t} - \frac{q}{qt-o}$ at these poles are $-q/j$, 1 and $-q/(q+j)$.

Rows and columns beyond $p + k$ vanish. Let $t \in [p + k, N)$ be a border node, so that $u(t) = 2$. Let t' be a node of the other kind. If $t' < k$, then $u(t') = 0$. The partial sum over $(t', t]$ then has valuation at least -1, and $t - t' \in (p, 2p)$ is prime to p. Otherwise $u(t') \ge 1$ and $v_p(M[t', t]) \ge -2$. In both cases the rescaled entry is divisible by p.

The remaining entries. For nodes below $p + k$ the only poles are k and p. Put $C_0 = [0, k)$ and $C_1 = [k, r)$. Recall that $\omega = q^{-2}$ is the normalisation of (2.1). The reduction has the following blocks. On $C_0 \times C_0$ it is the regular part plus x on the diagonal. On $C_0 \times C_1$ the entries are $-\omega q/(j(t - t'))$ with $t' \in C_0$ and $t \in C_1$. On $C_1 \times C_1$ it is the scalar matrix $\omega(q/j)^2\mathrm{Id}$. A border node $t \in [r, p)$ against $t' \in C_0$ gives $\omega q/(j(t - t'))$. A border node $t \in [p, p + k)$ against $t' \in C_0$ gives $\omega(1 - q/j)$ if $t' = t - p$, and 0 otherwise. All other border entries vanish.

Rank. Delete the zero rows and columns found above. This does not change the rank. Eliminate the scalar block on C_1. The Schur complement on C_0 is $\tilde{x}\mathrm{Id}$ plus a constant multiple of the matrix with entries $\sum_{i=1}^{p-r} \frac{1}{(t+i)(t'+i)}$ for $t, t' \in C_0$. Here $\tilde{x}$ is x plus a constant, and the computation uses the reflection $H_{p-1-n} \equiv H_n \pmod p$ of harmonic numbers. The border nodes $t \in [r, p)$ are indexed by $i = p - t$. They are all present because $N > p$. Even offsets are rows and odd offsets are columns, and together they cover $1 \le i \le p - r$. So row and column operations remove that multiple and leave $\left(\begin{smallmatrix} \tilde{x}\mathrm{Id} & \mathcal{J}_Y \\ \mathcal{J}_X & 0 \end{smallmatrix}\right)$. Here it is essential that x is an indeterminate, so that $\tilde{x}$ is a unit. A Schur complement on $\tilde{x}\mathrm{Id}$ then shows that the rank is $k + \mathrm{rank}(\mathcal{J}_X \mathcal{J}_Y)$. In $\mathcal{J}_X \mathcal{J}_Y$ the block between the high rows

and the high columns is zero, since their offsets $t-p$ have opposite parities. So $\operatorname{rank}(\mathcal{J}_X\mathcal{J}_Y) \le p-r$. In total the rank of the reduction is at most $(r-k)+k+(p-r)=p$, and the corank is at least $r+m-p$. Lemma 4.2 gives $v(\det) \ge (r+m-p)-\tau' = -(2(r-k)+3m)$. Deleting a common node changes the potentials by at most one. □

Under the hypothesis $N \le p+k$ the third pole does not occur, and the proof is shorter. The border nodes beyond $p+k$ only contribute rows and columns that vanish modulo p. This observation allows the whole range $N \le 2p$, which is automatic on the interval $r < p < r+m$.

4.3.3. *Pole-free windows: the bound (W).* Like the rank bounds, the bound (W) does not come from the valuations of the entries alone. Between two consecutive poles the step function V of (4.2) is constant. The reductions of the fibres of nearby residue classes are then translates of one another. Kernel vectors can be transported from one fibre to the next. Their pairings vanish modulo p^2 by Lemma 4.3. The corner bound (4.1) turns this into a saving at second order.

A *cut* is an integer t with $t \equiv 0$, $t \equiv k_p$ or $t \equiv r \pmod p$. A window $[t_1, t_2)$ of integers is *cut-free* if no cut lies in the open interval (t_1, t_2).

Theorem 4.7. *Assume that every pole below N is simple and $N < p^2$. Let $n \ge 0$ with $np \le m$, and put $N_n = r + 2(m-np)$. Let $[t_1, t_2)$ be a cut-free window with $\max(r, N_n - p) \le t_1 \le t_2 \le \min(N_n, r+p)$. Then every member satisfies*

$$e_p(r, I) \le 2r + 2m - (t_2 - t_1).$$

For $n = 0$ the windows lie in the border above $\max(r, N-p)$, and the bound applies to primes $p > m$. For $n = \lfloor m/p \rfloor \ge 1$ it applies to the small primes $p \le m$. For these primes only (T1) and (T1′) are available among the assignment bounds. Read modulo p, the window range is then the arc of residues that receive an odd number of border nodes. The saving over (T1′) is the amount, if any, by which the length of the longest cut-free run of this arc exceeds $\|m\|_p$. For other n the window range is empty. The proof uses the frame only through its poles. For a frame with several digamma terms the same statement holds when the cuts are the residues of all the poles and of r.

Proof. Let $\widetilde{\Phi} = p^2\Phi$, with independent diagonal variables on the common nodes. Every member is a coefficient of this determinant, so a bound for it bounds every member. Let $i < i'$ be two positions of the window. They lie in different residue classes. The translation $t \mapsto t + (i'-i)$ maps the nodes of the fibre of i bijectively onto those of the fibre of i'. It maps common nodes to common nodes, since no $t \equiv r$ lies in $(i, i']$. It maps border nodes to border nodes, since both classes lie in the window range. It maps X' to Y' when $i'-i$ is odd. There is no pole in $(i, i']$. So the step function V and the diagonal residues are unchanged, and so are the differences of nodes.

Hence the reduced block of the fibre of i' equals that of i when $i'-i$ is even, and its transpose when $i'-i$ is odd.

In the window the fibres alternate between one more row than column and one more column than row. Take a nonzero left kernel vector $\mathbf{y}$ of one block and transport it to every fibre of the window. It is a left kernel vector on the fibres with a surplus row and a right kernel vector on the others. Its moment pair is the same everywhere. By Lemma 4.3 all pairings between these vectors vanish modulo p^2. There are n_1 such vectors on rows and n_2 on columns, with $n_1+n_2=t_2-t_1$. Extending them to unimodular matrices, (4.1) gives $v(\det\widetilde{\Phi})\geq t_2-t_1$. Since $\det\widetilde{\Phi}=p^{2(r+m)}\det\Phi$, the theorem follows. □

4.3.4. *The rank bound (R).* Let $2m<p$. Write $r=k_0p+r_p$ with $0\leq r_p<p$. Note that $k_p=\lceil jp/q\rceil$, since $0<o<q$. Put $\mathrm{arc}_0=[0,k_p-1]$ and $\mathrm{arc}_1=[k_p,p-1]$. For $k\geq 0$ and $\iota\in\{0,1\}$ let

$$n_{k,\iota}=\big|[r,r+2m)\cap[kp+\min\mathrm{arc}_\iota,\ kp+\max\mathrm{arc}_\iota]\big|,$$
$$n_{\mathrm{sq}}=\big|[\max(0,r+2m-p),\min(r,k_p))\big|,$$

and

$$\Pi_R=2(r+m)-\max\big(m+n_{\mathrm{sq}},\ n_{0,0}+n_{1,0}+2n_{\mathrm{sq}},\ n_{k_0,0},\ n_{k_0,1},\ n_{k_0+1,0},\ n_{k_0+1,1}\big). \tag{4.4}$$

Proof of the bound (R). Every node is $t=i+pk$ with residue $i\in[0,p)$ and digit k. Write $k(i)$ for the number of common nodes in the fibre of i. The border has length $2m<p$. So this fibre contains the common digits $0,\dots,k(i)-1$ and at most one border digit, namely $k(i)\in\{k_0,k_0+1\}$. This border node is a row or a column according to the parity of $i+pk(i)-r$. The step function V is constant on the fibre up to the bit $\iota(i)=\mathbf{1}[i\geq k_p]$. Thus the reduced block of a fibre depends only on $(k(i),\iota(i))$ and on the role of its border digit. Its entries have the Löwner form of Lemma 4.4.

A fibre with a border row has a left kernel vector with nonzero moment pair. A fibre with a border column has such a right kernel vector. For the digit 0 below the pole the step function vanishes. So these fibres share the moment direction $(1,0)$. A fibre of type $(1,0)$ without border digit has the zero block and contributes one row and one column. Choosing one such vector per fibre kills $m+n_{\mathrm{sq}}$ rows and as many columns. This gives the saving $m+n_{\mathrm{sq}}$.

Within a class of fibres of the same type (k,ι) all moment pairs are equal. So by Lemma 4.3 the pairings between the killed rows and columns of one class vanish modulo p^2. Then (4.1) gives a saving equal to the number of killed coordinates of the class. This number is $n_{k,\iota}$ for $k\geq 1$ and $(k,\iota)\neq(1,0)$. It is $n_{0,0}+n_{1,0}+2n_{\mathrm{sq}}$ for the class of direction $(1,0)$. Taking the best class gives (4.4). A deleted common node changes at most one fibre. Give up that fibre, and one further fibre to restore the balance of rows and

columns. The trivial certificate then costs at most two killed coordinates of each kind. The smaller matrix compensates for them, so the debit is 0. □

4.3.5. *Separated border nodes: the bound (S).* The bound (S) uses a different source of cancellation. Choose a window $[t_0, t_v]$ of border nodes whose residues lie in one arc between consecutive poles. Consider a border node outside it. Suppose the interval between the node and t_0 contains a singular index. Then the residues modulo p of p times the partial sums at the two nodes differ, because two singular contributions in an interval of length less than p cannot cancel. The leading coefficient in x of the bordered determinant of a fibre is this residue. So the cross product of the two bordered fibre determinants has a nonzero leading coefficient. The border node is then eliminated against t_0 with no further information. Count the residues of the window together with the residues of the eliminated border nodes. This gives an explicit bound, a minimum over the four window types $(k, \iota) \in \{k_0, k_0 + 1\} \times \{0, 1\}$, which we now state.

Let $2m \le p$, and let k_0, r_p, arc_0, arc_1 and $n_{k,\iota}$ be as in §4.3.4. For a type (k, ι) let $\mathcal{K}_{k,\iota}$ be the interval of the integers $kp + i$ with $i \in \mathrm{arc}_\iota$, and with $i < r_p$ if $k = k_0 + 1$. The border nodes in $\mathcal{K}_{k,\iota}$ form the window, and there are $n_{k,\iota}$ of them. Let n_K be the number of border nodes in $\mathcal{K}_{k,\iota}$ and in its two translates $\mathcal{K}_{k,\iota} \pm p$. Put

$$n_\emptyset = p - |\mathcal{K}_{k,\iota}| - (2m - n_K), \qquad n_+ = (n_{k,\iota} - m)_+.$$

Since $2m \le p$, each residue class receives at most one border node. So n_K is the number of border nodes whose residues are those of $\mathcal{K}_{k,\iota}$. And $n_\emptyset$ is the number of the other residue classes that receive no border node. Then

$$\Pi_S = \min_{(k,\iota)} \big(2r + m - n_+ - (n_+ - n_\emptyset)_+\big), \tag{4.5}$$

where a type whose window is empty contributes $2r + m$. The bound (S) is $e_p(r, I) \le \Pi_S + |I|$. Its proof carries out the elimination above for every border node outside the window. We refer to the formal proof for it.

4.3.6. *The border wrapping around: the bound (U_n).* When $p \le 2m$ the border meets some residue classes more than once, and the bound (S) is not available. The bound (U_n) extends it to these primes. It works at third order in the Smith normal form. We show two things. After the elimination of a block of unit pivots, the rescaled matrix has the Smith invariants of a smaller geometry, to which the separation argument of (S) applies. The third-order information of that argument survives the elimination.

Write $m = np + \bar{m}$ with $0 \le \bar{m} < p$, so that $n = \lfloor m/p \rfloor$. Put

$$d = \|m\|_p = \min(\bar{m}, p - \bar{m}), \qquad \bar{r} = r + m - d.$$

For $i \in \{0, 1, 2\}$ and $\iota \in \{0, 1\}$ let $Z_i = p\lfloor r/p \rfloor + (n + i)p$, and put

$$w_0 = Z_i + \min \mathrm{arc}_\iota, \quad t_0 = \max(\bar{r}, w_0), \quad t_v = \min(Z_i + \max \mathrm{arc}_\iota,\, \bar{r} + p - 1).$$

Consider the virtual geometry with common block $[0, \bar{r})$ and border $[\bar{r}, \bar{r}+2d)$. In it, let $n_+ = \big(\min(t_v + 1, \bar{r} + 2d) - t_0 - d\big)_+$. Let n_K be the number of border nodes in $[w_0, t_v]$ and in its two translates by $\pm p$, and let $n_\emptyset = p - (t_v + 1 - w_0) - (2d - n_K)$. Then

$$\Pi_{U,n} = \min_{i,\iota}\Big(2(r+m) - d - n_+ - (n_+ - n_\emptyset)_+\Big), \tag{4.6}$$

all subtractions being truncated. This is the expression (4.5) for the virtual geometry $(\bar{r}, d)$, with the windows $[w_0, t_v]$ in place of $\mathcal{K}_{k,\iota}$. Note that $2(r + m) - d = 2\bar{r} + d$.

Theorem 4.8. *Let $p > p_0$ be a prime with $q(r + 2m) < p^2$, and let $n \geq 0$ with $np \leq m < (n + 1)p$. Assume $p \leq 2m$ if $n = 0$. Then every member satisfies $e_p(r, I) \leq \Pi_{U,n}$.*

We first describe the proof for $n = 0$, that is $m < p \leq 2m$. We then give the changes for $n \geq 1$. As in the proof of Theorem 4.7 we work with $\widetilde{\Phi} = p^2\Phi$. We put an independent indeterminate x_t on each common diagonal entry. We work over the discrete valuation ring $\mathcal{O}$ of Gauss valuations in these indeterminates. Every member is a coefficient of $\det \widetilde{\Phi}$, up to the deletion of common rows and columns, which we treat at the end.

Balanced pivots. For $n = 0$ there are $2m - p$ residue classes that carry two border nodes $t < t + p$, one a row and the other a column. There are $2d$ classes that carry one border node, and they lie in an interval of length $2d < p$. Eliminate the common block. Its fibres are invertible because the x_t are indeterminates. Consider a class with two border nodes. Let A be its common fibre, let $\mathbf{a}$ and $\mathbf{a}'$ be the couplings of its border column and border row to A, and let $\varkappa$ be its border entry. After the elimination the class has the pivot $\varpi = \varkappa - \mathbf{a}'^{\top} A^{-1} \mathbf{a}$. As the x_t tend to infinity, ϖ tends to a nonzero multiple of the residue of p times the increment of the frame over $(t, t+p]$. This interval contains exactly one pole of each digamma term. Let $k'p$ be the multiple of p in it, and t' the pole of the second term. If the two contributions cancelled, then $q(t' - k'p) \equiv o$ with $|q(t' - k'p) - o| < q$. Hence $q(t' - k'p) = o$, which is impossible for $0 < o < q$. So these $2m - p$ pivots are units.

After them, the matrix of size $r + m = \bar{r} + d$ is equivalent over $\mathcal{O}$ to $\operatorname{diag}(\mathrm{Id}_{\bar{r}}, p\,\Xi)$, with Ξ of size d. Its Schur complement on the border has the expansion

$$S = p\,S_1 + p^2(S_2 - \mathcal{C}_{UB}\mathcal{B}^{-1}\mathcal{C}_{BU}) + O(p^3). \tag{4.7}$$

Here $\mathcal{B}$ is the diagonal of the balanced pivots. The matrices $\mathcal{C}_{UB}$ and $\mathcal{C}_{BU}$ are the first-order couplings between these pivots and the single border nodes. The correction term in (4.7) is not negligible. The following identities account for it.

Lemma 4.9 (bordered inverses)**.** *Let A be an invertible square matrix over a commutative ring, and let $\mathbf{u}$ be a vector.*

(a) *Let A' be obtained from A by replacing one column by a vector $\mathbf{w}$, and suppose that A' is invertible. Let $\mathbf{u}'$ be obtained from $\mathbf{u}$ by replacing the corresponding entry by $\mathbf{u}^\top A^{-1}\mathbf{w}$. Then $\mathbf{u}'^\top A'^{-1}\mathbf{u} = \mathbf{u}^\top A^{-1}\mathbf{u}$.*

(b) *Suppose that the matrix*

$$A' = \begin{pmatrix} A & \mathbf{a} \\ \mathbf{a}'^\top & \varkappa \end{pmatrix}$$

is invertible. Then $(\mathbf{u}, y)^\top A'^{-1}(\mathbf{u}, \mathbf{a}'^\top A^{-1}\mathbf{u}) = \mathbf{u}^\top A^{-1}\mathbf{u}$ for every scalar y.

No symmetry of A or A' is assumed.

Proof. (a) Comparing columns gives $\mathbf{u}'^\top = \mathbf{u}^\top A^{-1}A'$. (b) We have $A'\,(A^{-1}\mathbf{u}, 0)^\top = (\mathbf{u}, \mathbf{a}'^\top A^{-1}\mathbf{u})^\top$. □

Moments. In the separation argument of (S), a border node is credited when its moment relative to a chosen reference node vanishes. That moment is a bilinear form $\mathbf{u}^\top A^{-1}\mathbf{u}$ in the inverse of the common fibre A. After the elimination, the common fibre of a class with two border nodes is replaced by a mixed block A' as in Lemma 4.9. Its lower node is a row and its upper node a column, or conversely. The added product in the inverse of A' is exactly the product subtracted in (4.7). Lemma 4.9 shows that the mixed moment equals the old one on every class that the separation argument credits. The single border nodes outside the window are separated from it as in (S), because two singular contributions in an interval of length $2d < p$ cannot cancel. The count of (S), applied to the virtual geometry $(\bar{r}, d)$, then gives (4.6). It uses the third-order analogue of (4.1). Suppose that n_+ kernel vectors on each side survive, and that at most n_0 residue classes carry a nonzero source. Then $\sum_i \min(v_i, 3) \geq d + n_+ + (n_+ - n_0)_+$, where $v_1, v_2, \ldots$ are the valuations of the Smith invariants. The first term d is the factor p of $p\,\Xi$.

Members. Deleting k common nodes lowers the trivial bound $2(r + m)$ by $2k$. Consider a class with one deleted common node and an unchanged direction. It keeps its moment. Indeed, let A be the common fibre and delete its last coordinate. The residual of the smaller solution is supported on that coordinate. The equality of the two directions forces its pairing with the load to vanish. A class whose direction changes gives a new separation instead. Classes with two or more deleted nodes, and classes outside the window, cost at most one each. Let k_1, k_2 and k' be the numbers of classes of these three kinds. Then $k_1 + 2k_2 + k' \leq k$. An elementary inequality for the count $n_+ + (n_+ - n_0)_+$ shows that the loss is at most $2k$. Hence the bound holds for every member with debit 0.

Larger n. Let $np \leq m < (n + 1)p$. Each class of the remaining border absorbs n pairs of border nodes. Its block after the elimination is a mixed block. Its rows and columns are the nodes of $2n + 1$ consecutive quotients $\beta, \beta+1, \ldots, \beta+2n$. Here β is the number of common nodes in the class, and the quotient of a node t in the class of a residue i is $(t - i)/p$. The window

counts are those of the first case at the scale $r + np$, translated by np. The new difficulty is to show that a remaining border node separated from the window by a pole has a nonzero coupling. The directions are now those of $(n+1) \times (n+1)$ bordered systems.

Lemma 4.10 (bordered systems and rational interpolation)**.** *Let $\mathbb{K}$ be a field and $\mathcal{Z} = \{0, 1, \dots, 2n\}$, with $(2n)! \neq 0$ in $\mathbb{K}$. Let $\mathcal{L} : \mathcal{Z} \to \mathbb{K}$, and let $\mathcal{Z} = X \sqcup Y$ with $|X| = n+1$ and $|Y| = n$. Consider the bordered system*

$$\sum_{y \in Y} \frac{\mathcal{L}(y) - \mathcal{L}(x)}{y - x}\, c_y + \vartheta = \mathcal{L}(x) \qquad (x \in X)$$

in the unknowns c_y and ϑ. Its solutions correspond bijectively to the pairs $(\mathcal{N}, \mathcal{D})$ of polynomials with $\mathcal{D}$ monic of degree n, $\deg \mathcal{N} \leq n$ and $\mathcal{N}(t) = \mathcal{L}(t)\mathcal{D}(t)$ for $t \in \mathcal{Z}$. Under this correspondence ϑ is the coefficient of t^n in $\mathcal{N}$. In particular the system is nonsingular exactly when the interpolation problem of type (n, n) has a unique solution. Then $\vartheta = \lim_{t \to \infty} \mathcal{N}(t)/\mathcal{D}(t)$ depends only on $\mathcal{L}$ and not on the split.

Proof. Let $\mathcal{W}(t) = \prod_{y \in Y} (t - y)$, and let $\mathcal{W}_y = \mathcal{W}/((t-y)\mathcal{W}'(y))$ be the Lagrange polynomials of Y. Given a solution, put $\mathcal{D} = \mathcal{W} - \sum_y c_y \mathcal{W}'(y)\, \mathcal{W}_y$ and $\mathcal{N} = \vartheta \mathcal{W} - \sum_y c_y \mathcal{L}(y) \mathcal{W}'(y)\, \mathcal{W}_y$. Then $\mathcal{N}(y) = \mathcal{L}(y)\mathcal{D}(y)$ for $y \in Y$. Dividing $\mathcal{N}(x) - \mathcal{L}(x)\mathcal{D}(x)$ by $\mathcal{W}(x) \neq 0$ gives the equation of the row x. Conversely, Lagrange interpolation on Y recovers $c_y = -\mathcal{D}(y)/\mathcal{W}'(y)$ and ϑ from $(\mathcal{N}, \mathcal{D})$. □

Lemma 4.11 (harmonic loads)**.** *Let $\mathcal{A} \in \mathbb{K}$ with $\mathcal{A} + i \neq 0$ for $0 \leq i < 2n$, and let $\mathcal{L}_{\mathcal{A}}(t) = \sum_{0 \leq i < t} 1/(\mathcal{A} + i)$. Let the characteristic of $\mathbb{K}$ be 0 or larger than $2n$. For every split, the bordered system of Lemma 4.10 with $\mathcal{L} = \mathcal{L}_{\mathcal{A}}$ is nonsingular. Its direction is $\vartheta = \mathcal{L}_{\mathcal{A}}(n) + \mathsf{c}_n$, with $\mathsf{c}_n \in \mathbb{K}$ depending only on n. Consequently, for the harmonic load $\mathcal{L}(t) = H_t$ on the quotients $\beta, \dots, \beta + 2n$ the direction is $H_{\beta+n} + \mathsf{c}_n$. The directions of two classes with β and $\beta + 1$ common nodes differ by $1/(\beta + n + 1)$.*

Proof. Write Δ for the forward difference. For $l \geq 1$ one has $\Delta^l \mathcal{L}_{\mathcal{A}}(k) = (-1)^{l-1}(l-1)!/(\mathcal{A} + k)_l$, where $(y)_l$ is the rising factorial. One also has $\Delta^l[\binom{t}{k} f(t)](0) = \binom{l}{k} \Delta^{l-k} f(k)$. By Lemma 4.10 a solution is a pair $(\mathcal{N}, \mathcal{D})$ with $\mathcal{N} = \mathcal{L}_{\mathcal{A}}\mathcal{D}$ on $\mathcal{Z}$. Since $\mathcal{N}$ has degree at most n, this means $\Delta^l(\mathcal{L}_{\mathcal{A}}\mathcal{D})(0) = 0$ for $n < l \leq 2n$. Write $\mathcal{D} = \sum_k \mathsf{d}_k \binom{t}{k}$ and insert the two formulas. After removing nonzero factors of the rows and columns, these n equations become a system for the coefficients d_k with $k < n$. Its matrix is the Cauchy matrix $\big(1/(l-k)\big)_{n < l \leq 2n,\, 0 \leq k < n}$, and its right-hand side comes from $\mathsf{d}_n = 1$. The Cauchy matrix is nonsingular. So $\mathcal{D}$ is unique, which is the nonsingularity.

The same computation shows that $\mathsf{d}_k/(\mathcal{A} + k)_{n-k}$ does not depend on $\mathcal{A}$. For $k < n$, $\Delta^{n-k} \mathcal{L}_{\mathcal{A}}(k)$ is a constant multiple of $1/(\mathcal{A} + k)_{n-k}$. Hence $\vartheta = \Delta^n(\mathcal{L}_{\mathcal{A}}\mathcal{D})(0)/n! = \mathcal{L}_{\mathcal{A}}(n) + \mathsf{c}_n$ with c_n independent of $\mathcal{A}$. For the harmonic

load on the quotients $\beta+t$ we have $H_{\beta+t} = H_\beta + \mathcal{L}_{\beta+1}(t)$. A constant added to $\mathcal{L}$ is added to ϑ. So the direction is $H_\beta + \mathcal{L}_{\beta+1}(n) + \mathsf{c}_n = H_{\beta+n} + \mathsf{c}_n$. □

Remark 4.12. The interpolation problem of Lemma 4.11 can be solved in closed form. Its denominator is the polynomial $\mathcal{D}_n(t) = \sum_{k=0}^{n} \binom{n}{k}\binom{2n-k}{n-k}(\mathcal{A}+k)_{n-k}\, t(t-1)\cdots(t-k+1)$, of Hahn type [19]. The constant is $\mathsf{c}_n = 2H_n$. The determinant of the bordered system is $\mathrm{sf}(n)^2\,\mathrm{sf}(n-1)^2/(\prod_{x\in X, y\in Y}|x-y|\cdot\prod_{i<n}(\mathcal{A}+i)_{n+1})$, with $\mathrm{sf}(n) = \prod_{i\le n} i!$. All its prime factors are at most $2n$ or divide some $\mathcal{A}+i$. These closed forms are not used in the proof and are not part of the formalization.

Separation for $n \ge 1$. Between a remaining border node and the reference node of the window lies a cut. Suppose first that the cut is a pole of the second digamma term. Deform the shift to $s+pY$, with an indeterminate Y. Evaluate at the value of Y at which that pole becomes central. After multiplication by the vanishing factor, the bordered systems of the two classes become block systems built from Cauchy matrices on the quotients. The square blocks are nonsingular by Cauchy's determinant formula. A rectangular Cauchy matrix with $k+1$ rows and k columns has a left kernel vector with nonzero coordinate sum. Otherwise the rational function $\sum_i y_i/(T_i-T)$, formed from a kernel vector (y_i) and the row nodes T_i, would vanish at the k column nodes while its numerator has degree less than k. This gives the two directions the values 0 and the weight of the term, which differ.

Suppose next that the cut is a multiple of p. Take instead the leading coefficient in Y. Every fractional pole term tends to 0, and the loads become the harmonic numbers. The two classes have β and $\beta+1$ common nodes. Lemma 4.11 gives directions that differ by $1/(\beta+n+1)$. This is a unit because $\beta+2n+1<p$, by $q(r+2m)<p^2$.

In both cases the comparison is made on the cleared numerators of the original determinants, not on inverses. The result returns to $Y=0$ by Gauss's lemma. The cleared denominators are products of the elementary denominators of the entries, which are units at $Y=0$. So a lower bound for the Gauss valuation of the deformed determinant gives the same lower bound for every coefficient of the original one. With these separations, the count and the treatment of the members are those of the case $n=0$.

4.4. **Profiles.** Let $\mathcal{E}$ be the set of integer expressions in r, p, m, k built from integer constants with the following operations: addition, multiplication by integer constants, truncated subtraction $(X-Y)_+$, min, max, and the distance $\|X\|_p$ of an expression X to $p\mathbb{Z}$. The quantities $p\lfloor r/p\rfloor$ and $r \bmod p$ are also allowed. Every bound of Theorem 4.5 lies in $\mathcal{E}$. For (R) and (S) this is (4.4) and (4.5). The ends of the intervals $\mathcal{K}_{k,\iota}$ are $kp = p\lfloor r/p\rfloor + (k-k_0)p$ plus 0, k_p, k_p-1, $p-1$ or r_p-1. For (U_n) this is (4.6) with $d = \|m\|_p$ and $Z_i = p\lfloor r/p\rfloor + (n+i)p$. For (W) the window ends are the cuts nearest to the ends of the window range, which are expressions of the same kind.

The *homogenisation* $\hat{\tau}(\theta)$ of $\tau \in \mathcal{E}$ replaces r, p, m, k by $1, \theta, \alpha, \kappa\theta$. It replaces every constant by 0, $\|X\|_p$ by $\theta\|\hat{X}/\theta\|$ and $p\lfloor r/p \rfloor$ by $\theta\lfloor 1/\theta \rfloor$. It is piecewise affine with rational breakpoints, finitely many in every interval bounded away from 0.

Lemma 4.13 (homogenisation)**.** *For each $\tau \in \mathcal{E}$ there is an explicit constant* $\mathrm{err}(\tau)$ *such that* $|\tau(r,p,m(r),k_p) - r\hat{\tau}(p/r)| \le \mathrm{err}(\tau)$ *for all* $r, p \ge 1$*. It is computed recursively from* $\mathrm{err}(m) = \frac{1}{2}$, $\mathrm{err}(k) = 1 - \frac{1}{q}$ *and the rules* $\mathrm{err}(\tau + \tau') = \mathrm{err}(\tau) + \mathrm{err}(\tau')$ *and* $\mathrm{err}(\min(\tau, \tau')) \le \max(\mathrm{err}(\tau), \mathrm{err}(\tau'))$.

Proof. The real version of τ, with the constants dropped, is positively homogeneous of degree one. Every operation is 1-Lipschitz in the maximum norm. The inputs differ from their homogeneous values by $|m - \alpha r| \le \frac{1}{2}$ and $|k_p - \kappa p| \le 1 - \frac{1}{q}$. Induct on τ. □

The conditions of Theorem 4.5 are linear inequalities in r, p, m, k, strict or not. They homogenise to intervals of θ. At an endpoint the error of such an inequality is a combination of $m - \alpha r$ and $k_p - \kappa p$. With the rounding $m = \lceil \alpha r - \frac{1}{2} \rceil$ two endpoints are *exact*. These are $\theta = 2\alpha$, from $2m < p$, and $\theta = (1 + 2\alpha)/(1 + \kappa)$, from $N \le p + k$. At an exact endpoint the condition holds for every prime on the correct side. The other endpoints are exact, or hold outside a strip of primes of bounded width around the endpoint. On that strip another bound must be used.

4.5. **The checker and the content theorem.** For the root class j let $\mathcal{P}_j(\theta)$ be the minimum of the homogenised bounds $\hat{\tau}(\theta)$ over the bounds τ of Theorem 4.5 valid at θ. A *content table* for the frame F_s and the geometry of parameter c consists of the following data. It contains the parameters (q, α, K). For every root class j it contains an increasing sequence $0 < \theta_0 < \theta_1 < \cdots < \theta_{\mathrm{end}}$. On each piece $(\theta_{i-1}, \theta_i]$ it contains an affine function $\ell_i(\theta)$ and a constant $\mathrm{err}_i \ge 0$. Each piece is tagged with a bound τ of Theorem 4.5. At each inexact endpoint of the validity interval of τ, the piece is also tagged with a second bound valid across that endpoint. Finally, the table contains a constant ℓ_0 for the prefix $(0, \theta_0]$ and the rational number Λ_{fin}. The *checker* accepts the table if the following conditions hold.

(C1) For each shift of the table, every unit class modulo the modulus q^* of the prime number theorem used is assigned a root class with a table.

(C2) On each piece ℓ_i is nonnegative, and the piece lies in the validity interval of its bound. Moreover $\hat{\tau}$ lies below ℓ_i at both ends and at both one-sided limits of each breakpoint inside the piece.

(C3) At each inexact endpoint the second bound is valid on both sides and lies below ℓ_i there. The constant err_i exceeds the error constants of Lemma 4.13, the debits times K, the strip widths times the slopes, and the trivial bound at the scales below an explicit threshold.

(C4) $\ell_0 \ge 2 + 2\alpha$ and $0 < \theta_0 \le 1$. Also $\theta_{\mathrm{end}} \ge \max(q(1 + 2\alpha), p_0/2)$ and $\kappa_j \theta_{\mathrm{end}} \ge 1 + 2\alpha$.

(C5) $\frac{1}{\varphi(q)}\sum_j\big(\ell_0\theta_0+\sum_i\int_{\theta_{i-1}}^{\theta_i}\ell_i(\theta)\,d\theta\big)=\Lambda_{\mathrm{fin}}$.
All of these are comparisons of rational numbers.

Theorem 4.14 (content)**.** *Let $0<c\le 1$, and let F_s be a frame with s of denominator q. Suppose that the checker accepts a content table of depth K with s among its shifts. Then for every even r and every member with $|I|\le K$,*

$$\log\operatorname{den}P_{r,I}\le(\Lambda_{\mathrm{fin}}+o(1))\,r^2.$$

Proof. By definition $\operatorname{den}P_{r,I}=\prod_p p^{e_p(r,I)}$. Fix a piece $(\theta_{i-1},\theta_i]$ of the root class j. Let p be a prime in a unit class assigned to j, with $\theta_{i-1}r<p\le\theta_i r$. Let r be beyond the threshold of (C3). Then the conditions of the tagged bound hold, either exactly or outside a strip where the second bound applies. Lemma 4.13 with (C2) and (C3) gives $e_p(r,I)\le r\,\ell_i(p/r)+\mathrm{err}_i$. Below the threshold the trivial bound is at most err_i.

For $p\le\theta_0 r$ the bound (T1) gives $e_p\le\ell_0 r+O(1)$. The primes with $p^2\le q(r+2m)$, the prime powers and the finitely many primes below p_0 contribute $O(r^{3/2}\log r)$. For $p>\theta_{\mathrm{end}}r$, (C4) gives $N\le p$ and $k_p\ge N$, and (T5) gives $e_p=0$.

Fix a unit class modulo q^*. By the prime number theorem in arithmetic progressions,

$$\sum_{\substack{\theta_{i-1}r<p\le\theta_i r\\ p\text{ in the class}}} r\,\ell_i(p/r)\log p=\frac{r^2}{\varphi(q^*)}\int_{\theta_{i-1}}^{\theta_i}\ell_i(\theta)\,d\theta+o(r^2).$$

The class j receives $\varphi(q^*)/\varphi(q)$ unit classes modulo q^*, because the map that sends a unit to qs times its inverse permutes the units modulo q. Summing and using (C5) gives the theorem. The constants err_i contribute $O(r)$. □

The tables of this paper are generated from the functions $\mathcal{P}_j$ themselves. The pieces are the maximal intervals on which $\mathcal{P}_j$ is affine and attained by one bound. So $\Lambda_{\mathrm{fin}}=\frac{1}{\varphi(q)}\sum_j\int\mathcal{P}_j$ up to the prefix. Every bound of Theorem 4.5 is available to the tables: (T1)–(T5), (LP1), (LP2), (R), (S), the windows (W) and the bounds (U_n). The windows with $n=0$ are used for $m<p$, between consecutive cuts.

On $(0,\alpha]$, that is for $p\le m$, the only other bounds are (T1), (T1′) and (U_n). With (T1′) alone the profile there is the sawtooth $2+2\alpha-\theta\|\alpha/\theta\|$. Its integral is

$$\int_0^{\alpha}\Big(2+2\alpha-\theta\Big\|\frac{\alpha}{\theta}\Big\|\Big)d\theta=2\alpha+\frac{\alpha^2}{2}+2\alpha^2\log 2,$$

since $\int_1^{\infty}\|y\|y^{-3}\,dy=\frac{3}{2}-2\log 2$. The windows with $n=\lfloor m/p\rfloor\ge 1$ lower this profile on the strips $\alpha/(n+1)<\theta\le\alpha/n$. The tables use them for $n\le 8$, again with each window running between two consecutive cuts. For $q=3,4,5,7$ the tables also use (U_0) on $(\alpha,2\alpha]$ and (U_n) on the strips with $1\le n\le 24$. The table for $q=2$ uses only the other bounds. Beyond the

strips used, a table resolves the sawtooth into teeth down to α/n_0. It ends with a constant piece on $(0, \alpha/n_0]$, which costs $\alpha^2 T_{n_0}$ with $0 < T_n \le 1/(4n^2)$. The number Λ_{fin} is then rational, and (C5) is an identity between rational numbers. The bounds depend on the shift s only through the root class, and their profiles only through (q, j, α). So one table serves all shifts with denominator q.

5. Nonvanishing

The deduction needs, for every rational x_0 and every large scale, one member of the family that does not vanish at x_0. For the frames of (2.1) this holds in characteristic zero. The number of deleted common nodes is bounded independently of r. This is the statement used in Section 8.

5.1. **Selection in characteristic zero.** Normalise the frame so that the divided differences of F at consecutive nodes have prefix values f_t, with $f_{t+1} - f_t = \Delta_F(t)$. The diagonal prefix $\tilde f_t$ satisfies $\tilde f_{t+1} - \tilde f_t = -\Delta_F'(t)$. For F_s,

$$\Delta_F(z) = \omega\Big(\frac{1}{z+1} - \frac{1}{z+s}\Big) = \frac{\omega(s-1)}{(z+1)(z+s)}.$$

For F_χ, with $\omega = -\frac{1}{6}$,

$$\Delta_F(z) = \frac{1-z}{216\,(z+1)(z+\frac{1}{3})(z+\frac{1}{2})}.$$

Theorem 5.1 (selection)**.** *Let F be a frame of (2.1), and consider the alternating geometry of parameter c. There is K, depending only on $|\mathcal{T}|$ and on the period 2 of the border, with the following property. For every large even r and every $x_0 \in \mathbb{Q}$ some member $P_{r,I}$ with $|I| \le K$ satisfies $P_{r,I}(x_0) \neq 0$.*

Sketch of proof. Write $N = r + 2m$. By barycentric interpolation, a vector in the right kernel of $\Phi_{r,\emptyset}(x_0)$ corresponds to a pair of polynomials (u, w) of degree less than $r + m$. The pair satisfies the *value conditions* $\mathsf{w}(t) = f_t \mathsf{u}(t)$ at the N nodes. It satisfies the *jet conditions* $\mathsf{w}'(t) = f_t \mathsf{u}'(t) + (x_0 - \tilde f_t)\mathsf{u}(t)$ at the common nodes t. The left kernel is described in the same way. Two such pairs have zero cross product $\mathsf{w}_1\mathsf{u}_2 - \mathsf{w}_2\mathsf{u}_1$. Indeed its degree is at most $2(r+m)-2$, while the value and jet conditions give $N + r = 2(r+m)$ zeros, counted with multiplicity.

Consider a pair whose degrees are small enough. Write $\Delta_F = n_\Delta / d_\Delta$ in lowest terms. The discrete Wronskian

$$\mathrm{Wr}(z) = d_\Delta(z)\big[\mathsf{w}(z+1)\mathsf{u}(z) - \mathsf{w}(z)\mathsf{u}(z+1)\big] - n_\Delta(z)\mathsf{u}(z)\mathsf{u}(z+1)$$

has more prescribed zeros at the nodes than its degree allows. So it vanishes identically. Then w/u would be a rational antidifference of Δ_F. This is impossible, because the simple pole of Δ_F at -1 is isolated in its class modulo $\mathbb{Z}$. The other poles of Δ_F are $-s$ for F_s, and $-\frac{1}{3}$ and $-\frac{1}{2}$ for F_χ. None of them is congruent to -1 modulo $\mathbb{Z}$. Clearing denominators

in $\mathsf{w}(z+1)/\mathsf{u}(z+1) - \mathsf{w}(z)/\mathsf{u}(z) = \Delta_F(z)$ would propagate a root of the denominator indefinitely in one direction. Counting the zeros along chains of nodes of step 2 bounds the dimension of the kernel. So the corank of $\Phi_{r,\emptyset}(x_0)$ is at most a number K that depends only on $|\mathcal{T}|$ and on the number of such chains. That number is bounded independently of r.

Finally, the left and right kernels map to the same interpolation space. At a common node t their coordinates are the values $\mathsf{u}(t)$ times nonzero factors. No nonzero kernel pair has u vanishing at every common node. Otherwise, dividing by the node polynomial of the common nodes would give a pair of degree less than m that satisfies the value conditions at all N nodes, and the Wronskian argument applies again. Hence there is a set I of $|I| =$ corank common nodes such that no nonzero left or right kernel vector vanishes on I. The rows outside I then span the row space, and the columns outside I span the column space. So the principal minor that deletes I is nonzero. □

The theorem applies to every frame of (2.1). For F_χ the pole of Δ_F at -1 is again isolated in its class modulo $\mathbb{Z}$, since the other poles are $-\frac{1}{3}$ and $-\frac{1}{2}$. In the formal proof the depth is $K \le 64$ for all the frames of this paper.

6. Analytic decay

In this section we bound $|P_{r,I}(x)|$ for x near ξ. The argument is that of OpenAI [31] for square determinants, with three changes. The determinant has two families of nodes, one for the rows and one for the columns. The comparison measure is a density on finitely many segments rather than an explicit equilibrium measure. The inequalities that remain are verified on a finite cover by cells. The result is one theorem for every geometry parameter $c \le 1$ and every frame of (2.1).

6.1. **A contour representation.** Hermite's formula gives $\mathsf{L}_F[t,t'] = \frac{1}{2\pi i}\oint F(z)\upsilon_t(z)\upsilon_{t'}(z)\,dz$ with $\upsilon_t(z) = 1/(z-t)$. The contour encloses the nodes and no pole of F. Write $F = \sum_{\varsigma\in\mathcal{T}} w_\varsigma \psi(z+\varsigma)$ as in §3. The reflection formula $\psi(w) = \psi(1-w) - \pi\cot\pi w$ splits $F = \tilde{F} + E$. Here $\tilde{F}$ is analytic in the left half-plane and $O(1/|z|)$ there, and

$$E(z) = -\pi \sum_{\varsigma\in\mathcal{T}} w_\varsigma \cot\pi(z+\varsigma).$$

The function E is 1-periodic. Since the weights sum to zero, $E(z) = O(e^{-2\pi|\operatorname{Im} z|})$ for $|\operatorname{Im} z| \ge 1$. For F_s one has $E(z) = \pi[\cot\pi(z+s) - \cot\pi z]$.

Let Γ be a polygonal path, symmetric under complex conjugation. It crosses the real axis at a point $x^* < 0$ that is not a pole of F, runs through the left half-plane and ends in two rays. The poles of F are the points $-(i+\varsigma)$ with $i \ge 0$ and $\varsigma \in \mathcal{T}$. Moving the contour to Γ picks up the residues of F at the poles in $(x^*, 0)$. The part of the integrand with $\tilde{F}$ integrates to 0 over Γ. Hence, for all integers $t, t' \ge 0$,

$$\mathsf{L}_F[t,t'] = \int \upsilon_t \upsilon_{t'}\, d\Omega. \tag{6.1}$$

Here Ω is the complex measure with the mass $-\operatorname{res}_y(F)$ at each pole $y \in (x^*, 0)$ of F and the density $(2\pi i)^{-1}E(z)\,dz$ on Γ. This is a Gram representation with respect to a measure with finitely many atoms and an exponentially small density on Γ. After scaling by r, the atoms lie on a segment $[-r_0, 0]$ with density $|\mathcal{T}|$. The separation condition on x^* is a finite check on the fractional parts of $\varsigma - x^*$ for $\varsigma \in \mathcal{T}$.

6.2. **Two-family Andréief bound.** For a set $C'' \subseteq C$ of common nodes put $\mathcal{M}_{C''} = \omega\, \mathsf{L}_F[C'' \sqcup X', C'' \sqcup Y']$. Consider a member with common nodes $C' = C \setminus I$. Put $R' = C' \sqcup X'$, $Q' = C' \sqcup Y'$ and $n_* = |C'| + m$. By multilinearity in the diagonal,

$$P_{r,I}(\xi + h) = \sum_{C'' \subseteq C'} h^{|C'|-|C''|} \det \mathcal{M}_{C''}. \tag{6.2}$$

Andréief's identity [14] for the two families $(\upsilon_t)_{t\in R'}$ and $(\upsilon_{t'})_{t'\in Q'}$ gives

$$\det \mathsf{L}_F[R', Q'] = \frac{1}{n_*!} \int \det[\upsilon_t(z_i)]\, \det[\upsilon_{t'}(z_i)] \prod_{i=1}^{n_*} d\Omega(z_i).$$

Cauchy's determinant formula evaluates both factors. We take absolute values. The logarithmic potential theory behind what follows is in [37]. We obtain

$$|\det \mathsf{L}_F[R', Q']| \le \frac{1}{n_*!} \int \prod_{i<i'} |z_i - z_{i'}|^2 \prod_{\mathcal{F}\in\{R',Q'\}} \frac{\prod_{t<t'\in\mathcal{F}} |t' - t|}{\prod_{i,\, t\in\mathcal{F}} |z_i - t|} \prod_i d|\Omega|(z_i).$$

The integrand is the exponential of minus a discrete logarithmic energy. The free points z_i repel each other with weight 1. Each node family repels itself with weight $\frac{1}{2}$ and attracts the free points with weight 1. The density $|E| \approx e^{-2\pi|\operatorname{Im} z|}$ acts as an external field. The rows and columns differ only in the interleaved border. So the two node families have the same macroscopic density. After scaling by r, both are

$$\nu_c = \mathrm{Leb}|_{[0,1]} + \tfrac{1}{2}\mathrm{Leb}|_{[1,1+c]},$$

of mass $|\nu_c| = 1 + c/2$.

6.3. **The energy certificate.** For a regularisation radius $\rho > 0$ let $k_\rho(z, w) = -\frac{1}{2}\log(|z - w|^2 + \rho^2)$. Then $k_\rho(z, w) \le -\log|z - w|$.

Lemma 6.1 (conditional positive definiteness)**.** *Let $z_i \in \mathbb{C}$ be finitely many points and y_i real numbers with $\sum_i y_i = 0$. Then $\sum_{i,j} y_i y_j k_\rho(z_i, z_j) \ge 0$.*

Proof. We use Frullani's integral $\log T = \int_0^\infty (e^{-u} - e^{-Tu})\,du/u$ for $T > 0$. Together with $\sum_i y_i = 0$ it gives $\sum_{i,j} y_i y_j(-\log(|z_i - z_j|^2 + \rho^2)) =$

$\int_0^\infty \frac{e^{-u\rho^2}}{u} \sum_{i,j} y_i y_j e^{-u|z_i - z_j|^2}\, du$. Gaussian kernels are positive definite. Indeed, expanding $e^{2u\langle z, z'\rangle}$ in powers and using the binomial theorem writes the inner sum as a convergent sum of squares. □

A *certificate measure* is a probability measure σ, symmetric under conjugation, carried by finitely many segments $\mathcal{S}_i \subset \Gamma \cup [-r_0, 0] \cup [0, 1]$ with rational endpoints. On each segment it has a density $\sigma_i(y)\, dy$ in the parameter $y \in [0, 1]$ of the segment. Here σ_i is a polynomial with rational coefficients, and $0 \le \sigma_i \le \bar{\sigma}_i$ for a rational $\bar{\sigma}_i$. We write $\Theta_\sigma = \sum_i \bar{\sigma}_i / |\mathcal{S}_i|$ for the sum of the bounds of its densities with respect to arclength.

We compare the free points and the deleted nodes with $|\nu_c|\sigma$ through Lemma 6.1. This bounds the free–free energy from below by a sum of one-point terms and the constant $|\nu_c|^2 \mathcal{I}(\sigma)$. Riemann-sum comparisons replace the node sums by the potential U^{ν_c} and the node self-energy by $\mathcal{I}(\nu_c)$, up to $O(r \log r)$. With the external field one arrives at the function

$$\Upsilon(z) = |\nu_c|\, U^\sigma(z) - U^{\nu_c}(z) + (1 - \varepsilon_0)\pi |\operatorname{Im} z|$$

for a small fixed $\varepsilon_0 > 0$. Here $U^\sigma(z) = \int \log \frac{1}{|z-w|}\, d\sigma(w)$ is the logarithmic potential of σ, and U^{ν_c} is that of ν_c. A *certificate* consists of σ and rational numbers Υ_Γ, Υ_0, J, $\mathcal{I}_\sigma$, $\mathcal{I}_\nu$ with the following properties.

(a) $\Upsilon \ge \Upsilon_\Gamma$ on Γ.
(b) $\Upsilon \ge \Upsilon_0$ on $[0, 1]$, the scaled position of the deleted common nodes.
(c) $J \le \int_{-r_0}^0 \min(\Upsilon(x) - \Upsilon_\Gamma, 0)\, dx$.
(d) $\mathcal{I}(\sigma) \le \mathcal{I}_\sigma$ and $\mathcal{I}(\nu_c) \ge \mathcal{I}_\nu$.

Theorem 6.2 (decay and window)**.** *Let $0 < c \le 1$. Let F be a frame of (2.1) whose poles are separated from x^*. Let $(\sigma, \Upsilon_\Gamma, \Upsilon_0, J, \mathcal{I}_\sigma, \mathcal{I}_\nu)$ be a certificate for Γ and ν_c. Let $\rho > 0$, $\delta_\rho \ge \pi\rho\,\Theta_\sigma$, $\delta_{\mathrm{m}} > 0$ and $\delta_J > 0$ be rational, and put*

$$\delta = 3|\nu_c|^2 \delta_\rho + 4|\nu_c|^2 \delta_{\mathrm{m}} + 2|\mathcal{T}|\, \delta_J.$$

Let D and γ be rational numbers with

$$D \le 2(|\nu_c| \Upsilon_\Gamma + |\mathcal{T}| J) - |\nu_c|^2 \mathcal{I}_\sigma + \mathcal{I}_\nu - \delta, \qquad \gamma \ge 2(\Upsilon_\Gamma - \Upsilon_0).$$

Then for every depth K, every member of the alternating geometry of parameter c and every real h with $|h| \le e^{-\gamma r}$,

$$|P_{r,I}(\xi + h)| \le \exp\bigl(-(D - o(1))r^2\bigr).$$

The loss δ comes from replacing σ by finitely many atoms in the comparison with the free points. The regularisation radius ρ of Lemma 6.1 and the mesh of the atoms lower Υ_Γ by $|\nu_c|(\delta_\rho + \delta_{\mathrm{m}})$. They raise $\mathcal{I}_\sigma$ by $\delta_\rho + 2\delta_{\mathrm{m}}$. Darboux sums for the integral in (c) lower J by δ_J. Inserted in the expression for D, these changes give δ. The tolerances δ_{m} and δ_J may be any positive rational numbers, since the atoms can be taken as fine as needed. In the certificates of this paper $\rho = 10^{-12}$ and $\delta < 10^{-9}$.

The node energy $\mathcal{I}(\nu_c)$ is a finite rational combination of the values $\mathcal{H}(y_k)$, where $\mathcal{H}(y) = \frac{1}{2} y^2 \log|y| - \frac{3}{4} y^2$ and the y_k are the rational differences of the

endpoints of the two pieces of ν_c. The lower bound $\mathcal{I}_\nu$ is checked with rational enclosures of the logarithms. These come from the series of artanh and a reduction to $[1,2)$. The energy $\mathcal{I}(\sigma)$ is a finite sum of pair energies of polynomial densities on segments. Each is a closed form of the type polynomial times logarithm plus arctangent, and is bounded in the same way. The conditions (a)–(c) are statements about Υ on sets of positive measure. They are proved on a cover by cells.

6.4. **The cover by cells.** Each of the conditions (a)–(c) is checked on a finite cover by cells with exact rational arithmetic. On a cell, the potential U^σ is split into the contribution of the segments of σ near the cell and the rest. A near segment is bounded by a Taylor polynomial in the cell variable with an explicit remainder. For a cell that meets the line of a segment beyond its end, the same bound is used for the segment with its orientation reversed. At a corner of Γ the two incident segments are treated together. Far segments are bounded by a multipole expansion with an explicit tail. The potential U^{ν_c} is an explicit elementary function, and it is bounded above by monotonicity on the cell.

For (a) the cells are intervals of the arclength on each edge of Γ. The bound is proved on a horizontal neighbourhood of the edge of fixed width. Beyond the last vertex the ray is handled by an analytic tail bound. For (b) the cells are intervals of $[0,1]$. For (c) the integrand is bounded above on each cell of $[-r_0,0]$ by a constant or by a chord. Then J is the sum of the integrals of these majorants.

Proposition 6.3 (certified constants)**.** *For each row of Table 2 there is a certificate measure σ on a polygon Γ, with a cover by cells accepted by the checks above, for which Theorem 6.2 gives the constants D and γ of the table.*

All inequalities are verified by the Lean kernel, as described in Section 11. The certificate data are the polygon, the density σ and the cells. They were computed with floating-point optimisation and interval arithmetic in Arb [18]. Lean checks them with rational arithmetic only. The data play no role in the proof beyond being accepted.

Remark 6.4 (how good a certificate can be). For an exponent B the certificate meets B exactly when $B(D-\Lambda)-\gamma \geq 0$. This quantity is a concave function of σ. Its maximum over all probability measures on $\Gamma \cup [-r_0,0] \cup [0,1]$ is attained at a measure σ^*. For this measure $|\nu_c|\sigma^*$ is the weighted equilibrium measure of the constrained energy problem on this set, with the external field $(1-\varepsilon_0)\pi|\operatorname{Im} z| - U^{\nu_c}$ and the upper constraint $|\mathcal{T}|\,dx$ on the segment [37]. Weak duality expresses the deficit of any certificate as a sum of nonnegative terms. The first is the energy of $2B|\nu_c|(\sigma-\sigma^*)$. This is the difference between $2B|\nu_c|\sigma$ and the optimal dual measure $2B|\nu_c|\sigma^*$. So a certificate close to the best possible one must be close to σ^* in energy. Then $\Upsilon - \Upsilon_\Gamma$ is a small ripple on the support of σ^*. This is why a pointwise

verification on a cover is needed. This remark is not used in the proof of Theorem 1.1 and is not part of the formalization.

6.5. **Node removal and the window.** The argument applies to every subset $C'' \subseteq C'$ of the common nodes. Removing common nodes turns them into points of the configuration that are not integrated over. Only common nodes can be removed. A node removed from one family only would leave an uncancelled term of weight $\frac{1}{2}$. By condition (b) one obtains

$$|\det \mathcal{M}_{C''}| \le \exp\bigl(-(D-o(1))r^2 + \gamma r\,|C' \setminus C''|\bigr),$$

uniformly in C''. Insert this into (6.2) with $|h|e^{\gamma r} \le 1$. The sum has at most $2^{|C'|} = e^{O(r)}$ terms, each at most $e^{-(D-o(1))r^2}$. This gives Theorem 6.2. The deletion of at most K common nodes in a member changes the energy by $O(r)$.

7. The pair of shifts $(\frac{1}{3}, \frac{2}{3})$

For the frame F_χ of (2.1) we use the alternating geometry of parameter $c = \frac{1}{2}$ at all even scales r. Then $\alpha = \frac{1}{4}$ and $m = \lceil r/4 - \frac{1}{2} \rceil$. The decay and the nonvanishing are given by the general Theorems 6.2 and 5.1. What changes is the content. The frame has three digamma terms. So at a prime p the increments $g(k)$ of §3 have poles in three residue classes. The local bounds of Section 4 then become expressions in two first poles instead of one. We first describe these poles. We then give the bounds that we use for F_χ, and finally the constants.

7.1. **The first poles.** Let $p > 3$ be a prime. The increment $g(k) = \frac{1}{3k-2} - \frac{1}{2k-1} + \frac{1}{6k}$ has a pole modulo p exactly when p divides k, $3k-2$ or $2k-1$. Let k_A and k_B be the least $k \ge 1$ with $p \mid 3k-2$ and with $p \mid 2k-1$. Then

$$k_B = \frac{p+1}{2}, \qquad k_A = \begin{cases} \frac{p+2}{3}, & p \equiv 1 \pmod 6, \\ \frac{2p+2}{3}, & p \equiv 5 \pmod 6. \end{cases}$$

The residues of $p\,g(k)$ at k_A and k_B are 1 and -1 if $p \equiv 1 \pmod 6$, and $\frac{1}{2}$ and -1 if $p \equiv 5 \pmod 6$. In the first case $k_A < k_B$ and the two residues cancel. So the partial sums $\sum_{k \le t} g(k)$ are p-integral for $t < k_A$, and again for $k_B \le t < p$.

Both first poles are affine in one index. Define k_p by $6k_p = jp + 5$ with $0 < j < 6$. Then $j = 1$ for $p \equiv 1 \pmod 6$ and $j = 5$ for $p \equiv 5 \pmod 6$, $k_p = \frac{j}{6}p + O(1)$, and

$$(k_A, k_B) = \begin{cases} (2k_p - 1,\ 3k_p - 2), & p \equiv 1 \pmod 6, \\ (2k_p - p - 1,\ 3k_p - 2p - 2), & p \equiv 5 \pmod 6. \end{cases}$$

Thus j plays the role of the root class of §4, with q replaced by 6.

7.2. **The local bounds.** Fix a prime $p > 6$ with $6N < p^2$, where $N = r + 2m$. Every pole below N is then simple. Call a node t *regular* if $t < \min(k_A, k_B)$, or if $p \equiv 1 \pmod 6$ and t is a border node with $k_B \le t < p$. At a regular node the partial sum of the increments up to t is p-integral. When t is a common node, its diagonal entry is p-integral too.

Proposition 7.1. *Every member of the alternating geometry of parameter $\frac{1}{2}$ for F_χ satisfies $e_p(r, I) \le \tau$ for each of the following τ.*

(i) The bounds (T1) and (T1′), namely $2(r + m)$ and $2r + 2m - \|m\|_p$.
(ii) If $N \le 2p$, the bound $2r + 2m - n_{\mathrm{reg}}$. Here n_{reg} is the number of pairs formed by a regular node t and a side on which t lies, row or column, such that $t + p$ is not a node of the other side.
(iii) If $N < \min(k_A, k_B)$, the bound 0.
(iv) The bound $2r + 2m - (t_2 - t_1)$ of Theorem 4.7, for every n and every window $[t_1, t_2)$ as there, with the cuts $t \equiv 0$, $t \equiv k_A$, $t \equiv k_B$ and $t \equiv r \pmod p$.

Proof. The bounds (i) are proved as in §4.3.1. For (ii) we apply Lemma 4.1 with the following potentials. At a node that is not regular, the potential is 1 on its row and on its column. At a regular node t, the potential on its row is 1 if $t + p$ is a column node, and 0 otherwise. The potential on its column is 1 if $t + p$ is a row node, and 0 otherwise. Their sum is $2(r + m) - n_{\mathrm{reg}}$, and they dominate the weights W. Indeed, a diagonal weight is 0 at a regular common node and at most 2 in general. Two nodes that are not congruent modulo p have weight at most 1, since all poles are simple. Their weight is 0 if both are regular. Two distinct congruent nodes have weight at most 2. Since $N \le 2p$ and regular nodes are smaller than p, at most one of them is regular. Suppose that the row node t of such an entry is regular. Then its column node is $t + p$, so the potential on the row of t is 1. The case of a regular column node is symmetric.

For (iii), the condition $N < \min(k_A, k_B)$ implies $N < p$. Then no index below N is a pole and no two nodes are congruent modulo p, so every entry is p-integral. For (iv), the proof of Theorem 4.7 uses the frame only through its poles. The step function V and the diagonal residues are constant between consecutive cuts. This is what the translation between the fibres of the window needs. □

The bounds (T2)–(T5), (LP1), (LP2), (R) and (S) of Theorem 4.5 are not used for F_χ.

7.3. **The content.** The checker of §4.5 and Theorem 4.14 apply to F_χ with q replaced by 6. The root class is $j \in \{1, 5\}$ as above, and the bounds are those of Proposition 7.1. These bounds are expressions of the class $\mathcal{E}$ in r, p, m, k_A, k_B. On each class of primes modulo 6 the substitution above makes them expressions in r, p, m, k_p. Their homogenisations are taken with $\kappa = j/6$. The proof of Theorem 4.14 uses the frame only through these

bounds and the root class. It goes through with the bound (iii) in place of (T5) beyond the end of the table.

The table of the formal proof starts at $\theta_0 = \frac{1}{68} = \alpha/17$. Below it the trivial bound $\ell_0 r$ with $\ell_0 = \frac{5}{2} = 2 + 2\alpha$ is used. On $(\alpha/17, \alpha]$ the table uses the bounds (i) and the windows (iv) with $1 \le n \le 16$. Beyond α it uses (i), (ii), the windows (iv) with $n = 0$ between consecutive cuts, and the bound (iii), which ends the table. The checker accepts the table for the two classes $j = 1, 5$. The content estimate therefore holds for every even r and every member with $|I| \le 64$, with the content constant

$$\Lambda_\chi = 4.09592776731\ldots$$

This is an explicit rational number, whose numerator and denominator have 117 and 116 digits.

7.4. **Decay and nonvanishing.** The decay estimate is Theorem 6.2 for the frame F_χ, with $|\mathcal{T}| = 3$ and $c = \frac{1}{2}$. The node measure $\nu_{1/2}$ has mass $\frac{5}{4}$ and energy

$$\mathcal{I}(\nu_{1/2}) = \frac{75}{32} - \frac{9}{8}\log\frac{3}{2} - \frac{1}{16}\log 2.$$

A certificate measure on a polygon, with a cover by cells accepted by the checks of §6.4, gives

$$D = \frac{4643511791526209475773}{10^{21}} = 4.64351\ldots,$$
$$\gamma = \frac{37521870783566737}{5\cdot 10^{15}} = 7.50437\ldots$$

Nonvanishing is Theorem 5.1 for F_χ. Its increment Δ_F has its pole at -1 isolated in its class modulo $\mathbb{Z}$, and the depth is $K \le 64$. The deduction of Section 8, at the even scales, gives

$$\mu\big(\Psi(\tfrac{1}{3}, \tfrac{2}{3})\big) \le \frac{\gamma}{D - \Lambda_\chi} = 13.704516\ldots < 13.7046,$$

since $\Psi(\frac{1}{3}, \frac{2}{3}) = 36\,\xi_\chi$.

8. Proof of Theorem 1.1

The three estimates combine as follows.

Theorem 8.1 (deduction)**.** *Let F be a frame of (2.1) with its geometry. Let K be a depth such that, for every large r and every rational x_0, some member $P_{r,I}$ with $|I| \le K$ satisfies $P_{r,I}(x_0) \ne 0$, as in Theorem 5.1. Suppose that the content constant Λ of Theorem 4.14 and the constants D, γ of Theorem 6.2 satisfy $\Lambda < D$. Then $\mu(\xi) \le \gamma/(D - \Lambda)$.*

Proof. Let $\eta = D - \Lambda$, and let $\mu_0 > \mu_1 > \gamma/\eta$. Suppose that $|\xi - a/b| < b^{-\mu_0}$ with b large. Let r be the least admissible scale with $r \ge \mu_1 \log b/\gamma$. By Theorem 5.1 with $x_0 = a/b$ there is a member with $P_{r,I}(a/b) \ne 0$. By Lemma 3.4, $|\operatorname{den}(P_{r,I})\, P_{r,I}(a/b)| \ge b^{-r}$. Write $a/b = \xi + h$. Then

$|h| < b^{-\mu_0} \le e^{-\gamma r}$, because $\gamma r \le \mu_1 \log b + O(1) < \mu_0 \log b$ for large b. By Theorems 4.14 and 6.2,

$$|\operatorname{den}(P_{r,I})\, P_{r,I}(a/b)| \le e^{(\Lambda+o(1))r^2} e^{-(D-o(1))r^2} = e^{-(\eta-o(1))r^2}.$$

Comparing the two bounds gives $(\eta - o(1))r^2 \le r \log b$. But $\eta r \ge (\mu_1\eta/\gamma)\log b$ with $\mu_1\eta/\gamma > 1$. This is a contradiction for large b. □

Proof of Theorem 1.1. For each row of Table 1 the frame, the geometry parameter and the constants are those of Table 2. The content constant is given by Theorem 4.14 with the content table of the formal proof. For the pair $(\frac{1}{3}, \frac{2}{3})$ it is given by Section 7. The decay constants are given by Proposition 6.3, and nonvanishing by Theorem 5.1. In every row $\gamma/(D-\Lambda) < B$. Theorem 8.1, applied at the even scales, gives $\mu(\xi) \le B$. By (2.1), ξ is a nonzero rational multiple of $\Psi(s,t)$. This proves Theorem 1.1. □

(s,t)	c	Λ	D	γ	γ/η	B
$(\frac{k}{4}, 1)$	$\frac{1}{5}$	3.5149036	3.7447213	7.1904273	31.28751	31.2876
$(\frac{k}{3}, 1)$	$\frac{3}{10}$	3.3104458	4.0291249	6.8272548	9.499725	9.4998
$(\frac{k}{5}, 1)$	$\frac{2}{9}$	3.5128488	3.8135797	7.1655663	23.82717	23.8272
$(\frac{k}{7}, 1)$	$\frac{1}{6}$	3.6391158643	3.6424792669	7.4259546075	2207.870	2207.8697
$(\frac{1}{2}, 1)$	$\frac{3}{10}$	2.9923824	3.9785713	6.4350874	6.525207	6.5253
$(\frac{1}{3}, \frac{2}{3})$	$\frac{1}{2}$	4.0959278	4.6435117	7.5043742	13.70452	13.7046

TABLE 2. The data of the proof of Theorem 1.1, with k prime to the denominator. For each pair of shifts the table gives the geometry parameter c, the content constant Λ, the decay constant D, the window constant γ and the exponent. The constants are the rational numbers of the formal proof, rounded outward to the digits shown: Λ and γ up, D down. The column γ/η, with $\eta = D - \Lambda$, is the exact quotient, rounded up.

Remark 8.2 (relation to linear independence criteria). The deduction is a one-dimensional version of the familiar scheme behind the criteria of Nesterenko [29] and their use by Ball and Rivoal [2]. The nonzero integer $b^{|C\setminus I|}\operatorname{den}(P_{r,I})\, P_{r,I}(a/b)$ plays the role of a small linear form with integer coefficients. There are two differences. The lower bound for the form comes from an arithmetic nonvanishing statement rather than from an analytic asymptotic. The size r of the determinant is chosen in terms of b.

9. THE FAMILIES

We collect the facts specific to each row of Table 2. Section 7 treats the pair $(\frac{1}{3}, \frac{2}{3})$. In every other row the frame is F_s of (2.1) with $s = k/q$, $\omega = q^{-2}$

and $\xi_s = -q^{-2}\Psi(\frac{k}{q}, 1)$. The content table, the certificate measure and the cover are the data of the formal proof.

9.1. **The shifts $\frac{1}{4}$ and $\frac{3}{4}$.** Here $\xi_{1/4} = -\frac{1}{16}\Psi(\frac{1}{4}, 1)$ and $\xi_{3/4} = -\frac{1}{16}\Psi(\frac{3}{4}, 1)$. The geometry parameter is $c = \frac{1}{5}$, so $\alpha = \frac{1}{10}$. The two shifts $s = \frac{k}{4}$ with $k = 1, 3$ have root classes $j \equiv kp^{-1} \pmod 4$. These exchange the two classes of primes modulo 4. By the transport of §4.5 the two shifts have the same content constant. The same certificate serves both, since the decay depends on the shift only through the separation of the poles from the contour.

9.2. **The shifts $\frac{1}{3}$ and $\frac{2}{3}$.** Here $\xi_{1/3} = -\frac{1}{9}\Psi(\frac{1}{3}, 1)$ and $\xi_{2/3} = -\frac{1}{9}\Psi(\frac{2}{3}, 1)$. The geometry parameter is $c = \frac{3}{10}$. The content table uses the primes in the classes modulo 6. The two root classes $j = 1, 2$ correspond to first poles near $p/3$ and $2p/3$.

9.3. **The shifts $\frac{k}{5}$.** For $1 \le k \le 4$ the geometry parameter is $c = \frac{2}{9}$. The four shifts share one content table, indexed by the four root classes, and one certificate.

9.4. **The shifts $\frac{k}{7}$.** For $1 \le k \le 6$ the geometry parameter is $c = \frac{1}{6}$, and there are six root classes. Here $D - \Lambda$ is small, about $3.4 \cdot 10^{-3}$. The exponent is correspondingly large.

9.5. **The shift $\frac{1}{2}$.** Here $q = 2$, $\omega = \frac{1}{4}$ and $\xi_{1/2} = -\frac{1}{4}\Psi(\frac{1}{2}, 1)$. There is one root class, $j = 1$, with first pole $k_p = \frac{p+1}{2}$. The geometry parameter is $c = \frac{3}{10}$. As explained after Theorem 1.1, the resulting bound is weaker than the known ones for this number.

10. Limits of the method

The exponents are determined by three constants through $\gamma/(D - \Lambda)$. Since $\partial\mu/\partial D = -\gamma/\eta^2$, the exponent is most sensitive to the decay where η is small. For the shifts $\frac{1}{4}$ and $\frac{3}{4}$, $\gamma/\eta^2 \approx 136$. For the shifts $\frac{k}{7}$ it is about $6.6 \cdot 10^5$.

On the arithmetic side, the profiles $\mathcal{P}_j$ are minima of the bounds of Theorem 4.5. Every further local bound lowers Λ by the integral of the amount by which it lies below $\mathcal{P}_j$. The content constant of the method is the integral of the true exponent e_p, which the local bounds approximate from above. At third order this approximation is close to exhaustion for the shifts $\frac{k}{5}$ and $\frac{k}{7}$. We computed the Smith normal forms exactly at scales $r \le 2400$ and primes $p \le 2m$. There the minimum of the local bounds, with (U_n) included, equals the true exponent at all but a few isolated primes. For the shifts $\frac{k}{7}$ the invariants of valuation at least 4 contribute about $5 \cdot 10^{-6}$ to Λ. The bounds (U_n) gain about $7 \cdot 10^{-4}$. So the content of the method at third order is essentially used up. These computations are not part of the proof.

On the analytic side, Remark 6.4 explains why a certificate of the present form cannot do better than the constrained equilibrium measure on the

chosen contour. A larger decay constant needs a different contour or a different treatment of the energy problem. The geometry parameter c trades the two sides against each other. A longer border increases D, through the larger node measure. It also increases Λ, through the border rows and columns of the assignment bound.

11. Formalization

The proof of Theorem 1.1 is formalized in Lean 4 [10] with Mathlib [39], together with every theorem, proposition and lemma it uses. The remarks and Section 10 are not part of the formalization. The development is available at

https://github.com/slitsey/catalan-irrationality-measure

It combines analytic and arithmetic proofs with generated rational data. Kernel checks verify the content tables of Section 4 and the decay certificates of Section 6. Numerical computations alone are not used as proof.

The statements of Theorem 1.1 are in the file `ChallengeHybrid.lean`. Their proofs are in `SolutionHybrid.lean`, in the namespace `TrigammaDifferences`. Each final theorem concerns the classical number of one row of Table 1, written as a series as in Section 1. It states that for every $e > 0$ there is b_0 such that $b^{-(B+e)} < |\xi - a/b|$ for all $b \geq b_0$ and all $a \in \mathbb{Z}$ coprime to b. The theorems are the following.

- `lambda_plus_measure` and `lambda_minus_measure`: the pairs $(\frac{1}{4}, 1)$ and $(\frac{3}{4}, 1)$, with $B = 31.2876$.
- `chi3_measure`: the pair $(\frac{1}{3}, \frac{2}{3})$, with $B = 13.7046$.
- `chi3_plus_pi2_measure` and `chi3_minus_pi2_measure`: the pairs $(\frac{1}{3}, 1)$ and $(\frac{2}{3}, 1)$, with $B = 9.4998$.
- `fifths1_measure`, ..., `fifths4_measure`: the pairs $(\frac{k}{5}, 1)$ for $k = 1, \ldots, 4$, with $B = 23.8272$.
- `sevenths1_measure`, ..., `sevenths6_measure`: the pairs $(\frac{k}{7}, 1)$ for $k = 1, \ldots, 6$, with $B = 2207.8697$.
- `pi2_measure`: the pair $(\frac{1}{2}, 1)$, with $B = 6.5253$.

The shared results are in the namespace `CatalanCore.Uniform`.

- `Content.BankShared.Check.hasScaledContent_of_check` is Theorem 4.14. The local bounds of Theorem 4.5 form its catalogue, in `Check/Catalogue.lean`, `Check/LP2W.lean`, `Check/Prefix.lean` and `Check/UF.lean`, with the closed rank bound and the branches of Theorem 4.8. Its form for frames with several first poles is `Content.BankShared.KTerm.hasScaledContent_of_kcheck`.
- `Content.BankShared.TwoTerm.LP2HypW.gle` is Theorem 4.6.
- `Content.BankShared.TwoTerm.Pre.prefix_gle` is Theorem 4.7.

- `Content.UFSimple.WrapCheck.catValid_wrap` and `Content.UFSimple.WrapPrefixCheck.catValid_prefix_full` are Theorem 4.8 for $n = 0$ and for $n \geq 1$. Lemma 4.11 is `Content.UFSimple.WrapPrefixClosed.HSplit.det_ne` with `HSplit.dir`.
- `BlockGeom.hasNV0` is Theorem 5.1.
- `Decay.AltDecay.altGeom_hasScaledDecay` is Theorem 6.2. It uses the node energy check `Decay/Nodes/EnergyCheck.lean`, the contour check `Decay/ContourCheck.lean` and the cell checks `Decay/PolyWire3`.
- `measure_of_family_nv0_even` is Theorem 8.1 at the even scales.

The solution has no `sorry` and no `native_decide`. The command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The prime number theorem in arithmetic progressions enters the content asymptotics. Its formal source is the PNT+ project [40], through OpenAI's workspace and compatibility additions. We use its Wiener–Ikehara theorem.

The proofs were checked with Lean `v4.35.0-rc2` and the matching Mathlib release. The vendored files from OpenAI's repository and from the PNT+ project were converted to Lean's module system and ported to the current toolchain. The repository's README and provenance documents describe the changes.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author. The author posed the problems, directed the work and is responsible for this paper. The systems were Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. OpenAI's irrationality proof for Catalan's constant and its Lean formalization [31] underlie this work. We use Lean 4 [10], Mathlib [39], the PNT+ project [40] and Arb [18]. Arb, through `python-flint`, provides the interval arithmetic used to generate certificate data, which Lean re-checks. The formal verification in Lean is intended as an independent check of every step of the argument above.

Acknowledgements. We thank the authors of Mathlib and of the PNT+ project, and OpenAI for making its formalization available.

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